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SURVEY

Enhancing Autonomous System Security and Resilience With Generative AI: A Comprehensive Survey

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ABSTRACT This survey explores the transformative role of Generative Artificial Intelligence (GenAI) in enhancing the trustworthiness, reliability, and security of autonomous systems such as Unmanned Aerial Vehicles (UAVs), self-driving cars, and robotic arms. As edge robots become increasingly integrated into daily life and critical infrastructure, the complexity and connectivity of these systems introduce formidable challenges in ensuring security, resilience, and safety. GenAI advances from mere data interpretation to autonomously generating new data, proving critical in complex, context-aware environments like edge robotics. Our survey delves into the impact of GenAI technologies—including Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Transformer-based models, and Large Language Models (LLMs)—on cybersecurity, decision-making, and the development of resilient architectures. We categorize existing research to highlight how these technologies address operational challenges and innovate predictive maintenance, anomaly detection, and adaptive threat response. Our comprehensive analysis distinguishes this work from existing reviews by mapping out the applications, challenges, and technological advancements of GenAI and their impact on creating secure frameworks for autonomous systems. We discuss significant challenges and future directions for integrating these technologies within security frameworks to address the evolving landscape of cyber-physical threats, underscoring the potential of GenAI to make autonomous systems more adaptive, secure, and efficient.

INDEX TERMS GenerativeAI, artificial intelligence, autonomous systems, security, UxV.

I. INTRODUCTION

Autonomous systems, including self-driving cars, UxV, and robotic arms, are increasingly prevalent across various industries such as transportation, manufacturing, healthcare, surveillance, and agriculture. The market size for humanoid robots is anticipated to reach USD\$13.8 billion by 2028 [1]. These systems face multiple challenges, including operational safety, data integrity, and dynamic environmental interactions. Ensuring their security and resilience is crucial to prevent economic losses, protect human lives, and maintain public trust in these emerging technologies. For instance, autonomous vehicles are susceptible to security threats, with potential risks of unauthorized control by hackers [2],

posing significant safety risks to passengers [3]. Moreover, incorporating machine learning algorithms in Autonomous Vehicles (AVs) and robots enhance their perception capabilities through real-time processing of high-dimensional, multi-modal data from onboard sensors like cameras and LIDAR. However, these components may harbor hidden cybersecurity vulnerabilities, potentially exposing AVs to malicious attacks that can deviate their paths and endanger human life and property [4]. Additionally, the complexity of these challenges escalates notably in scenarios involving swarms of cooperating devices, as evidenced by incidents of autonomous taxis causing traffic disruptions.¹

This survey focuses on the critical role of Generative Artificial Intelligence (GenAI) in enhancing autonomous

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¹<https://edition.cnn.com/2023/08/14/business/driverless-cars-san-francisco-cruise/index.html> Accessed:02-08-2024

systems—a domain where precision, reliability, and adaptability are paramount [5], [6]. GenAI marks a pivotal advancement in artificial intelligence research, transitioning from merely understanding data to autonomously generating new data that aligns with learned patterns and distributions. As these technologies evolve, they increasingly influence various fields, fundamentally changing how machines interact, learn, and adapt to their environments. GenAI offers promising solutions by augmenting decision-making capabilities, refining models with synthetic data, optimizing predictive maintenance, and fortifying security measures. Leveraging models such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and LLMs, these systems can attain unparalleled levels of autonomy and resilience. However, integrating GenAI into autonomous technologies encounters substantial obstacles, such as data scarcity, adaptability to fluctuating conditions, and mitigation against sophisticated threats.

This paper demonstrates that GenAI significantly enhances capabilities in security features such as intrusion detection [7], risk assessment [8], and system resilience. The use of GenAI in autonomous technologies, such as UxV and edge robotics, which require advanced, continuous, and context-aware monitoring and management, is still in its early stages. This survey categorizes existing research to underscore GenAI's impact on cybersecurity, decision-making processes, and the development of resilient architectures. We focus on individual edge devices like robots and the aggregated information from the entire fleet. We explore the technological foundations and novel applications of GenAI in scenarios such as **Cybersecurity Enhancements** in UxV operations, **Predictive Maintenance** for autonomous vehicles, **Adaptive Traffic Management** systems, and **Robotic Automation**. These applications demonstrate how technologies such as GANs, VAEs, Transformer-based models, and LLMs not only address operational challenges at the individual level but also enhance fleet-wide intelligence through innovative approaches for predictive maintenance, anomaly detection, and adaptive threat response. We also explore how GenAI addresses the inherent challenges of autonomous systems and drives innovations that enhance their functionality and safety, enabling more complex, adaptive, and resilient technologies.

While previous works have explored traditional Artificial Intelligence (AI)/Machine Learning (ML) methods for autonomous systems, this survey uniquely focuses on the application of GenAI to enhance the security and resilience of these systems within cybersecurity frameworks, emphasizing its capability for dynamic content creation and real-time decision-making, crucial for addressing evolving cyber-physical threats [9], [10]. GenAI significantly enhances intrusion detection, risk assessment, and system resilience capabilities—where traditional approaches often fall short. The main contributions of this paper are: i) We demonstrate the role of GenAI in enhancing the security and resilience of autonomous systems through advanced data augmentation

and simulation and enhanced perception and decision-making. Specific case studies where these techniques have been effectively implemented are detailed, showcasing practical applications and real-world impact; ii) We map and analyze the capabilities of GenAI across diverse applications such as cybersecurity enhancements, predictive maintenance, adaptive traffic management, and robotic automation. This comprehensive evaluation illustrates how GenAI leverages complex algorithms to address operational challenges, significantly improving both individual unit performance and fleet-wide intelligence; iii) We address the main challenges of applying GenAI to autonomous systems and propose future directions, emphasizing the need for scalable solutions, integration of transparency, and enhanced robustness against attacks. These initiatives pave the way for resilient, ethical, and effective autonomous systems, setting a new standard for deploying GenAI technologies.

The remainder of the paper is structured as follows: Section II reviews and discusses previous relevant works. Section III serves as a primer on GenAI models and their fundamental principles, offering insights into the underlying mechanisms driving these technologies. In Section IV, we delve into the core of the paper, exploring the diverse applications of GenAI within autonomous systems. Section V addresses the challenges and outlines future research directions in this burgeoning field. Finally, in Section VI, we synthesize our findings and underscore the importance of continuous innovation in advancing the capabilities and security of autonomous systems empowered by GenAI. Figure 1 provides an overview of the paper's structure, which comprises four main sections.

II. RELATED WORKS

The field of Generative Artificial Intelligence (GenAI) has seen significant expansion, prompting a comprehensive review of peer-reviewed surveys to highlight its progress and pinpoint areas for further research. Dube [11] contributes by exploring the application of LLM in information security, organizing the discussion around two main themes: “security of AI” and “AI for security.” The first theme addresses adversarial attacks and defenses within AI systems, while the second investigates the potential of LLM to enhance the security of non-AI IT systems. Despite providing an in-depth analysis of penetration testing, vulnerability assessments, and threat intelligence, the study lacks an examination of LLMs in autonomous systems.

In addition, Zhu et al. [12] outlines the security risks associated with GenAI, such as vulnerabilities to attacks and the possibility of exploitation by adversaries. The paper introduces innovative defensive strategies, including “AI firewalls,” and advocates for revising regulatory frameworks to keep pace with swift technological developments and ensure the responsible integration of GenAI in critical sectors. This viewpoint is supported by other surveys assessing LLMs's impact on strengthening and potentially undermining

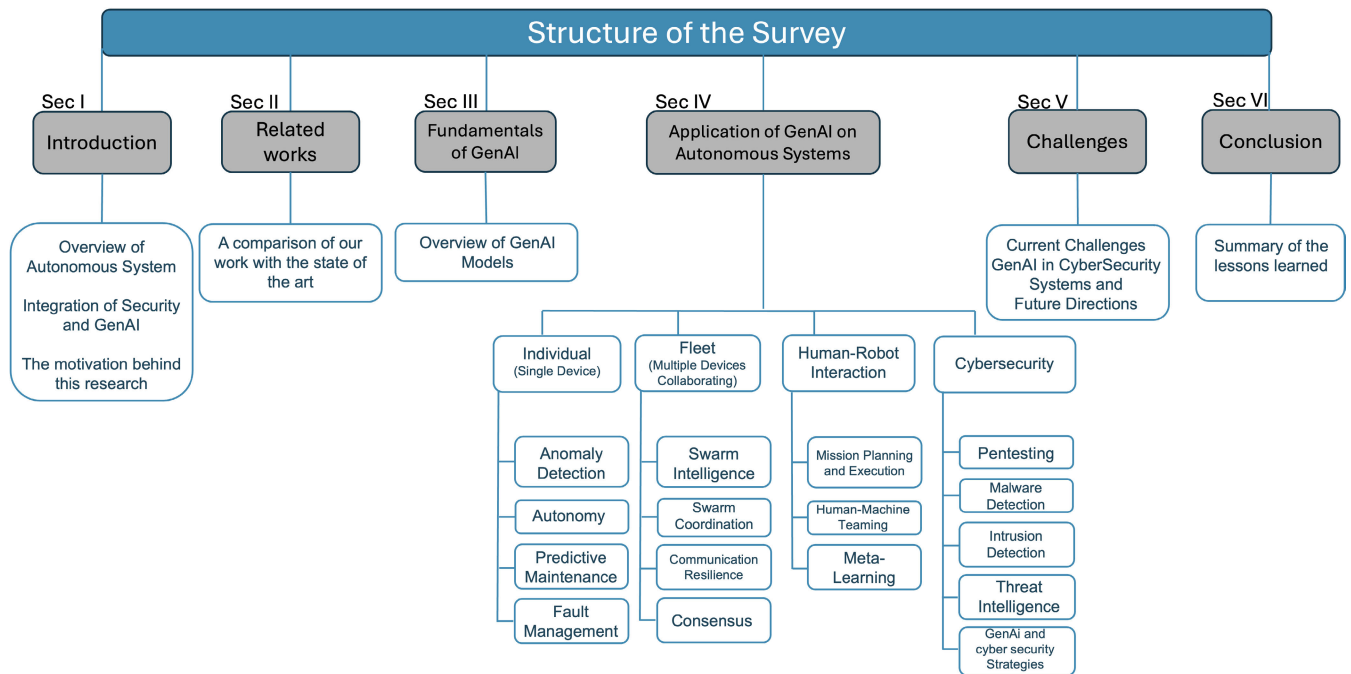


FIGURE 1. Taxonomy of the survey structure outlining the progression from fundamental concepts to specific applications and challenges of generative AI in autonomous systems.

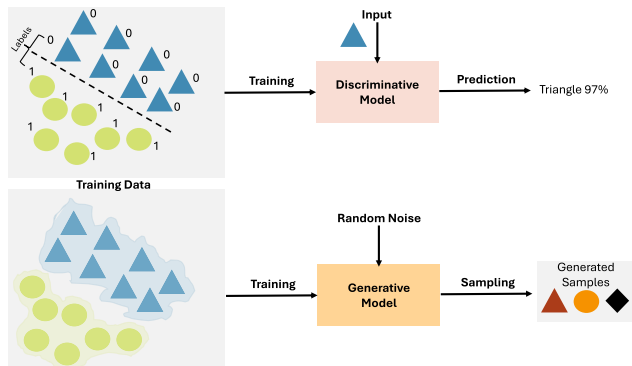


FIGURE 2. Difference between discriminative and generative models. A discriminative model learns decision boundaries for direct prediction, while a generative model learns data distributions for generating new samples.

cybersecurity. For instance, one study evaluates how LLMs enhances threat detection across different scenarios, while another suggests incorporating LLMs into threat intelligence and vulnerability assessments, promoting a sophisticated Mixture-of-Experts (MoEs) framework to broaden the use of LLM in cybersecurity tasks [13], [14].

Further exploration in the literature categorizes the influence of LLMs, such as ChatGPT and Bard, into positive applications, offensive uses, and vulnerabilities—termed “The Good,” “The Bad,” and “The Ugly.” This classification emphasizes the dual role of LLMs in cybersecurity, ranging from improving code security and data privacy to their potential exploitation in user-level attacks, underscoring the importance of research into model and parameter extraction attacks and the safe tuning of instructions [16]. Additionally,

Golda et al. [20] examines the privacy and security challenges introduced by GenAI in various sectors, including healthcare and business, advocating for collaborative solutions to promote ethical data practices and demanding more transparent regulations concerning user-generated content and model interpretability. Moreover, Munir et al. [27] presents a zero-trust framework specifically tailored to bolster security within power grid supply chains, illustrating effective management of GenAI-induced cyber threats through domain-specific GANs and ensemble learning for detecting synthetic identities.

Using LLMs in security is gaining significant attention. Ferrag et al. [17] extensively reviews the use of GenAI and LLMs in enhancing cybersecurity, covering applications across multiple domains such as hardware design security, intrusion detection, software engineering, design verification, cyber threat intelligence, malware detection, and phishing and spam detection. Their work analyzes 42 specific LLM models and highlights LLM vulnerabilities and mitigation strategies, assessing the performance and lifecycle of cybersecurity datasets. Zhang et al. [18] focuses on constructing cybersecurity-specific LLMs and their applications in various tasks like threat intelligence, vulnerability detection, secure code generation, and anomaly detection, and addresses existing challenges such as adversarial attacks and the need for robust evaluation metrics. Techniques like continual pre-training and supervised fine-tuning are highlighted, along with proposed future research directions to improve LLM integration in cybersecurity. Hassanin and Moustafa [19] reviews the evolution of LLMs, emphasizing their capabilities in automating threat intelligence, enhancing vulnerability

TABLE 1. Overview of the related work.

Reference	GenAI	LLM	Security	Autonomous Systems	Summary
Dube [11]	×	✓	✓	×	Focuses on LLMs in information security, discussing adversarial attacks and defense mechanisms.
Zhu et al. [12]	✓	×	✓	×	Examines security challenges of GenAI, proposing AI firewalls and updating threat management strategies.
Chen et al. [13]	×	✓	✓	×	Details how LLMs optimize threat detection and improve security operations in IT systems.
da Silva and Westphall [14]	×	✓	✓	×	Discusses LLM integration into threat intelligence and vulnerability assessments, highlighting the need for empirical validation.
Motlagh et al. [15]	×	✓	✓	×	Describes LLM's defensive and adversarial uses in the attack lifecycle, from reconnaissance to defense evasion.
Yao et al. [16]	×	✓	✓	×	Categorizes impacts of LLMs into beneficial, offensive, and vulnerable aspects within cybersecurity.
Ferrag et al. [17]	×	✓	✓	×	Analyzes 42 specific LLM models and highlights LLM vulnerabilities and mitigation strategies, assessing the performance and lifecycle of cybersecurity datasets.
Zhang et al. [18]	×	✓	✓	×	Analyzes the construction and application of cybersecurity-specific LLMs, addressing challenges like adversarial attacks and proposing future research directions.
Hassanin and Moustafa [19]	×	✓	✓	×	Discusses the evolution of LLMs for cybersecurity tasks, emphasizing automation, privacy preservation, and future research to enhance transparency and interpretability.
Golda et al. [20]	✓	×	✓	×	Discusses privacy and security challenges in GenAI, suggesting solutions involving multiple stakeholders.
Javaid et al. [21]	✓	×	×	✓	Reviews the integration of LLMs with UAV, emphasizing their potential to enhance UAV communication and operations. It addresses challenges such as computational demands, system integration difficulties, and latency issues in cloud processing.
Liu et al. [22]	✓	×	×	✓	Focuses on the use of GenAI in UAV swarm management, highlighting feature extraction and transformation capabilities.
Zeng et al. [23]	×	✓	×	✓	Explores the use of LLMs in robotics, emphasizing their impact on enhancing intelligence, interaction, and decision-making in robotic systems.
Kim et al. [24]	×	✓	×	✓	Explores the impact of LLMs on robotics, addressing their applications in communication, perception, planning, and control. It highlights challenges such as ensuring safety, the unpredictability of emergent abilities, and the lack of systematic guidelines for LLM integration in robotics.
Wang et al. [25]	×	✓	×	✓	Investigates the integration of LLMs and MLLMs in robotic tasks, focusing on the advanced reasoning and language comprehension capabilities for task planning.
Cao [1]	×	✓	×	✓	Explores the evolution of humanoid robots driven by GenAI, and LLMs towards interactive, real-time applications, highlighting the ongoing challenges in achieving human-like intelligence.
Neupane et al. [26]	×	×	✓	✓	Survey on security in AI-enhanced robotics, detailing vulnerabilities, attack surfaces, and ethical and legal concerns, while offering a taxonomy for understanding and mitigating these issues.
Our Work	✓	✓	✓	✓	Explores how GenAI and LLMs enhance the resilience, security, and safety of autonomous systems.

assessments, improving intrusion detection and network traffic analysis, and employing privacy-preserving techniques like differential privacy and federated learning. The paper also discusses using LLMs for realistic cybersecurity training scenarios, addressing adversarial attacks, and integrating LLMs with other AI techniques for transparency and interoperability. Additionally, the impact of LLMs on automating security operations, reducing human error, and increasing efficiency through Security Orchestration, Automation, and Response (SOAR) and integration into Security Operations Centers (SOCs) are highlighted. However, significant challenges remain, including LLM vulnerabilities to adversarial attacks, data poisoning, and DDoS attacks; robust evaluation metrics and comprehensive datasets; integration with other AI techniques; transparency and interpretability issues; privacy concerns; and the efficiency and scalability of automated security operations.

Zeng et al. [23] Kim et al. [25] Wang et al. [24] discuss the transformative impact of LLMs, while Cao [1] explore the role of GenAI and Large multimodal models (LMMs) in the field of robotics. Zeng et al. focus on how LLMs enhance robotics through natural language processing improvements, bolstering robot intelligence, autonomy, and human-robot interaction. They cover the application of LLMs in areas like control, perception, decision-making, and path planning, detailing the development, benefits, and technical approaches of LLMs in robotics, alongside potential challenges. Cao, on the other hand, details the evolution of humanoid robots influenced by these technologies, pushing towards more interactive, real-time, and multimodal applications. Despite these advancements, they emphasize the ongoing gap in achieving human-like intelligence, highlighting both the progress and the significant challenges in creating truly intelligent and humane humanoid robots.

Moreover, Wang et al. [25] explore the integration of LLMs and multimodal LLMs into robotic tasks, emphasizing their advanced reasoning and language comprehension capabilities for formulating action plans based on natural language instructions. They identify a critical gap in the compatibility of text-only LLMs with robotic visual perception, proposing using GPT-4V, the last MLLM from Open-AI, to bridge this gap and enhance embodied task planning. The surveyed literature highlights a need for more research on integrating LLMs with other modalities and planning algorithms to improve the efficiency and precision of robotic tasks. Additionally, Kim et al. [24] focus on integrating LLMs with intelligent robots, addressing language-conditioned behavior and control challenges. The authors review different approaches, such as image-based behavioral cloning and offline reinforcement learning, and emphasize the importance of developing methods that enable robots to comprehend and execute natural language instructions effectively.

Some works focused on exploring GenAI and LLMs applications in autonomous systems, specifically UAVs. Javaid et al. [21] analyze how different LLMs enhance individual UAV operations, emphasizing their roles in improving communication and operational efficiency. The authors present a detailed comparison of various LLMs and their specific utilities in UAV contexts, offering valuable insights for optimizing UAV capabilities. Liu et al. [22] explore generative models like GANs and VAEs in UAVs swarms, highlighting their ability to generate realistic training simulations crucial for robust system training. The paper presents a detailed approach to discussing the stability and reliability of GenAI models in generating diverse training scenarios, which is critical for the adaptability and efficiency of UAV swarms in dynamic environments.

Neupane et al. [26] provide a comprehensive review of the security landscape in AI-enhanced robotics. It highlights the critical areas of attack surfaces, ethical and legal concerns, and human-robot interaction security, presenting a taxonomy that categorizes potential threats and mitigation strategies. The paper emphasizes the vulnerabilities of AI-Robotics systems to various types of security threats, including physical attacks on hardware, digital attacks affecting software and data integrity, and other relevant attacks such as backdoors and malware. It also addresses the ethical and legal implications of robotic autonomy, focusing on issues like accountability, privacy, and the psychological impact of robots on human interactions.

Comparatively, our work extends beyond the specific applications discussed in these studies to provide a comprehensive overview of GenAI's impacts across a broader spectrum of autonomous systems, including UAVs, self-driving cars, and robotic arms. By categorizing existing research and highlighting the multifaceted applications of GenAI in cybersecurity, decision-making, and resilient architectures, our survey underscores the role of GenAI in advancing autonomous systems' trustworthiness, reliability, and security. This broader perspective not only integrates

the insights offered by the studies mentioned above but also sets a foundation for future explorations into the potential of GenAI to address emerging challenges and drive innovation in autonomous system technologies. Table 1 provides a comparative analysis of related works alongside our contribution, explaining their respective focus areas.

III. FUNDAMENTALS OF GENERATIVE AI

Generative Artificial Intelligence (GenAI) represents a significant breakthrough in AI, enabled by its capacity for content generation. This advancement is driven by generative models that analyze and interpret the distribution patterns in a dataset. The primary aim of GenAI models is to precisely capture and interpret the underlying probability distributions that define the dataset [28]. Through a deep comprehension of these distributions, generative models can produce new data instances that align with the original dataset's characteristics. This feature sets GenAI apart from traditional discriminative models, which are tailored to predict outcomes for unseen data through conditional probability. As shown in Figure 2, GenAI models create new data by learning from input data distribution, in contrast to discriminative models, which primarily focus on identifying class boundaries. Discriminative models excel in various tasks, such as classification and regression, by effectively defining the decision boundaries within a dataset for different classes or categories.

The field of GenAI is rapidly evolving, driven by a variety of generative methods such as transformers, Large Language Model (LLM), Generative Adversarial Networks (GAN), Variational Autoencoders (VAE), Normalizing Flow Model (NFM), and Generative Diffusion Model (GDM). These technologies find widespread application across various domains, from drone swarms and intrusion detection to communication security and mobile networks [29]. Innovations like ChatGPT [30], which are built on Generative Pre-Trained Transformer (GPT) models, have popularized GenAI by utilizing LLMs to produce coherent and contextually relevant accurate responses. LLMs utilize large datasets and Deep Learning (DL) technologies [31], particularly transformer architectures, to generate human-like content across various topics. Beyond textual output, generative models such as DALL-E and Stable Diffusion demonstrate the capability of AI to create visual content from textual or visual inputs, categorized under Vision Language Model (VLM). Additionally, the development of Multimodal Large Language Model (MLLM) integrates these capabilities, enabling models to simultaneously process and generate information across different data types—text, images, and audio [32]. This adaptability has broadened the use of LLMs in content creation, code generation, and translations, among other applications. Advanced models like GPT-4 [33], Gemini [34], Llama 2/3 [35], [36], Mistral [37], Falcon [38], Orca [39], and Claude [40] have been fine-tuned to generalize across a range of tasks, thereby enhancing the safety, security, and resilience of autonomous systems.

Over the past decade, there has been notable progress in AI models that automate the creation of efficient representations for various data types. Although LLMs often attract public attention, other generative AI models like GAN, VAE, and GDM also play essential roles. While less famous, these models are crucial in advancing research and development that improve drone security, safety, and resilience. The versatility of these generative techniques is encapsulated in Table 2, which summarizes the applications of each model, demonstrating their impact across various domains.

Initially proposed for translating between languages, transformer models excel at managing long-range dependencies in unstructured data through their key feature: the self-attention mechanism [28]. As depicted in Fig 3.d, this mechanism enables each embedding to attend to other sequences' embeddings, capturing intricate word relationships and contextual nuances. Prior methods required converting raw text into vector representations through a separate model. Transformers integrate these processes, allowing for the simultaneous handling of translation and embedding creation, which is pivotal for processing large volumes of unlabeled text and developing extensive language models with billions of parameters. Subsequent innovations have expanded transformer applications to include document summarization, question answering, code generation, in-depth analysis, intrusion detection, malware identification, and the translation of control instructions across robotic arms. While transformer models offer the significant advantage of extracting context from complex datasets, they also present challenges such as potential output inaccuracies, hallucinations, longer training times, slower inference rates, higher computational demands, and larger model sizes than earlier techniques.

Variational Autoencoders (VAEs) are generative models that leverage probabilistic techniques to encode input data into a compressed representation and decode it back into the original data space. Central to its architecture is encoding data into a latent space with a predefined prior distribution, typically Gaussian, which helps regulate the encoded data [41]. The encoding process uses the reparameterization trick, which allows gradients to flow through the random part of the model by expressing the latent variable as a deterministic function of the input and some random noise, making the training via stochastic gradient descent feasible. An illustration of the VAE's architecture is presented in Fig 3.a. The network consists of two main parts: the encoder, which maps the input data to the latent space, and the decoder, which reconstructs the data from this space. This probabilistic approach, underpinned by variational inference, allows VAE to approximate the posterior distribution of latent variables efficiently. It estimates this distribution by optimizing a lower bound on the likelihood of the data, often referred to as the ELBO (Evidence Lower Bound). The ELBO integrates two critical components into the VAEs loss function: the reconstruction loss and the KL divergence. The

reconstruction loss encourages the decoded samples to match the original inputs, enhancing the fidelity of output relative to the input. The KL divergence acts as a regularization term, promoting the latent variables to mimic the prior distribution, which is critical in ensuring that the learned latent space does not deviate significantly from the prior assumptions. This structure enables VAEs to effectively handle missing data and model the underlying distribution, making them particularly useful for anomaly detection, data augmentation, and image generation, where nuanced data variability is crucial.

Generative Adversarial Networks (GANs) consist of two neural networks in a dynamic competition: a generator that creates synthetic data and a discriminator that evaluates the authenticity of this data [42]. As depicted in Figure 3.b, the generator aims to produce increasingly realistic data, such as images, while the discriminator learns to distinguish between real and generated content. This interplay enhances the generator's ability to produce credible outputs that can deceive the discriminator, balanced by a game-theoretic equilibrium where both networks continuously adapt and improve. The generator and discriminator employ specific loss functions—binary cross-entropy for the discriminator and a loss function aimed at fooling the discriminator for the generator—to guide this adversarial training. Initially developed for image generation, GANs are now being applied across various domains, including text generation, drug discovery, video game content creation, and the enhancement of low-resolution medical images, demonstrating their versatility and broad utility. This makes them valuable for creating realistic synthetic datasets that can improve the development of autonomous systems and cybersecurity measures. However, challenges like mode collapse, where the generator produces limited types of outputs, catastrophic forgetting, where a network rapidly loses previously learned information, and stability issues in reaching convergence remain significant hurdles.

Generative Diffusion Models (GDMs) systematically add noise to data samples such as images and then methodically removes it to reconstruct the underlying data structure [43]. This process refines the original data and introduces slight variations that can generate new data points. Such capability has positioned GDM as a popular choice in image generation, enhancing the diversity and quality of datasets. Importantly, employing GDM for synthetic data generation is a growing trend that has been shown to facilitate the training of robust visual representations [44]. This technique is particularly useful in improving signal classification within autonomous systems. Despite its benefits, GDM tends to have longer sampling times and demands more complex architectures compared to GANs and VAEs, which may pose challenges in its application.

Normalizing Flow Models (NFM) are a class of models known for transforming simple, easy-to-understand probability distributions into more complex ones through a series of reversible transformations – each step meticulously

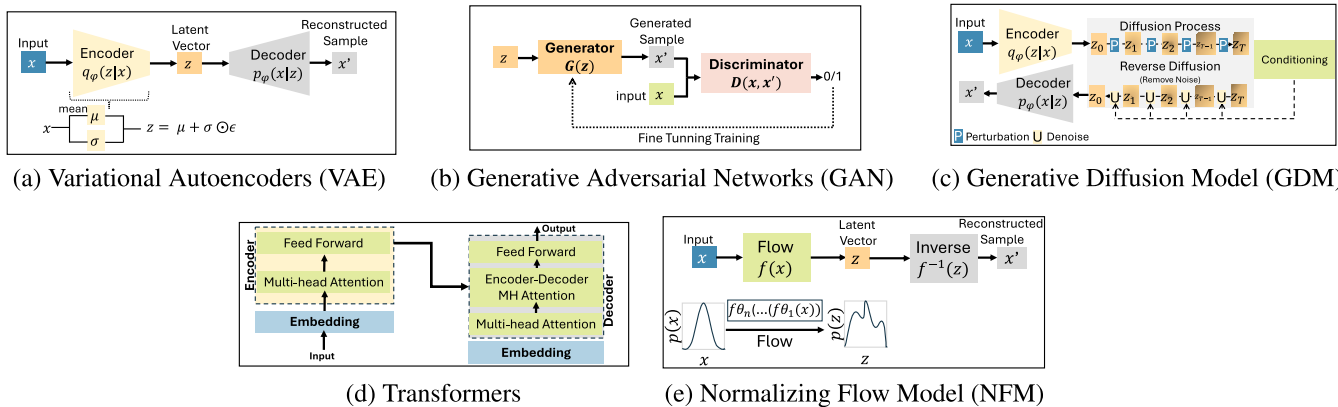


FIGURE 3. Visual representations of five key Generative Artificial Intelligence (GenAI) models.

TABLE 2. GenAI techniques and their applications.

GenAI Techniques	Applications
Transformers: Process sequential data using self-attention mechanisms to capture complex dependencies	Advanced state estimation for threat detection
LLMs: Process extensive textual data to generate nuanced, human-like responses	Decision-making in autonomous driving
GANs: Create realistic data by pitting two neural networks against each other in a feedback loop	Simulation data for robotic systems
VAEs: Generate new samples by learning efficient data encodings in a probabilistic framework	Fault detection
GDMs: Iteratively refine noise into structured data, mimicking the natural data generation process	Enhance the realism of simulation environments
NFMs: Transform data into complex distributions for advanced generative tasks	Anomaly detection
VLMs: Process both visual and textual inputs to understand and respond to natural language instructions	Navigation and interaction in autonomous systems

structured so that reversing the process is always an option [45]. This progression from simplicity to complexity is visualized in Fig 3.e. These models, often depicted as flows, start with a basic distribution like a normal distribution and gradually evolve into a distribution resembling the more intricate structure of the original data. This ability enhances classification and encoding schemes and allows the handling of more sophisticated data manipulations. The flexibility of NFMs allows them to adapt to various tasks, yet this adaptability can sometimes lead to inefficiencies. The training process involves adapting the model to transform the probability distribution of a dataset into a complex, reversible form, often requiring more computational resources and time than other generative techniques like GANs and VAEs. Concerning their application in anomaly detection and other areas, it is crucial to understand the inductive biases inherent in NFMs. These biases, while beneficial in guiding neural networks away from overfitting, can limit their effectiveness in anomaly detection [46].

IV. APPLICATION OF GenAI ON AUTONOMOUS SYSTEMS

An autonomous system must independently determine if it meets the performance criteria set by the user for a specific mission. The proliferation of autonomous technology across various sectors brings challenges across individual, fleet, human control, and cybersecurity domains. As these systems become increasingly complex and integrated into multiple domains, developing innovative approaches that can revolutionize their trustworthiness and reliability is

crucial to ensure their security, resilience, and safety. Many examples are being explored outside the specific UAVs and autonomous systems domain. However, they all have important implications for the future of UAVs and other autonomous systems. It is also important to note that many of the research projects described below are early in their maturity and are included here to flavor what might be possible as GenAI tools and techniques evolve.

A. INDIVIDUAL ENTITIES

GenAI demonstrates significant potential in enhancing security architectures, crucial for bolstering the security, resilience, and safety of autonomous systems such as drones, self-driving cars, robots, and embedded systems. Research in this area is exploring applications that include improving self-awareness, anomaly detection, autonomous driving, predictive maintenance, fault management, self-healing, and safe landing capabilities.

Self-awareness of autonomous systems is essential for comprehending and recognizing their operational state and functionality in various environments. This capability is key to ensuring drone health and accurately capturing and interpreting their current state. Such assessments cover hardware conditions, application performance, physical location, and security posture. However, practical challenges include motion blur in video feeds, erratic Global Positioning System (GPS) data, uncalibrated inertial guidance from Inertial Measurement Unit (IMU) sensors, and inconsistencies or gaps in internal monitoring data. The main goal here

is to process this noisy and inconsistent data effectively to precisely evaluate the drone's current state, which is crucial for obstacle detection and navigation. Accurate state estimation is imperative for autonomous decision-making, linking raw data streams to the drone's position, velocity, and orientation to ensure reliable performance in dynamic and complex settings [5]. Perception is vital for robots, enabling decision-making, action planning, and navigation in the real world [47]. Research in LLM-based robotic perception focuses on integrating language skills with sensory modalities—visual, auditory, and haptic—to enhance environmental understanding through passive and active behaviors. Visual modalities leverage pre-trained visual-language models for object recognition and scene description tasks. Auditory perception uses audio-language models to integrate sound with visual data for richer environmental understanding. Haptic perception interprets contact information, blending tactile and visual data to enhance tactile feedback. The section also differentiates between passive perception, which involves receiving data without actively seeking it, and active perception, which entails proactive information gathering to improve environmental interaction. This dynamic interplay of modalities and behaviors underpins advancements in robotic systems, enabling more sophisticated and context-aware operations [24].

Additionally, GenAI significantly aids in augmenting incomplete data and amalgamating information from diverse sources, thereby enhancing the reliability of the state estimation process [22]. This comprehensive approach enables drones to function effectively, even under challenging conditions. Wu et al. [48] discuss the integration of LLM and VLMs into robotics, highlighting significant safety and robustness challenges. The authors demonstrate how these integrations are susceptible to adversarial attacks that can manipulate robot actions, presenting severe safety risks.

The development of GenAI algorithms such as GANs, VAEs, along with traditional ML algorithms like Long Short-Term Memory (LSTM), addresses these data gaps and enhances vehicle safety, fault detection, predictive maintenance, and safe landing protocols. For instance, novel GAN techniques help in filling missing data and simplifying the integration of data streams for improved state assessments [49], assist in correlating internal log data with acoustic analyses [50], and pinpoint potential mechanical faults [51]. Researchers have also developed methods using Conditional Generative Adversarial Networks (CGANs) to estimate state variables for individual drones and drone swarms [52]. Table 3 summarizes the principal contributions of GenAI to individual autonomous systems.

1) ANOMALY DETECTION

Enhancing the analysis of drone sensor data to pinpoint abnormal conditions is critical for maintaining drone health and operational dependability. VAEs are particularly effective in boosting fault detection and isolation, capable of

identifying early indicators of system stress, which is vital for setting predictive maintenance schedules. Typically, machine learning classification algorithms are trained on categories such as faulty and not faulty, but datasets often lack adequate examples of rare conditions, like faults, which are uncommon in both public datasets and real-world applications. Chirayil Nandakumar et al. [53] provides a comprehensive examination of anomaly detection literature, particularly in Autonomous Robotic Missions (ARMs), and proposes a novel classification of anomalies based on spatial, temporal, and spatiotemporal characteristics.

Generative Adversarial Networks (GAN) are instrumental in synthesizing these rare categories, thus augmenting the training data available for machine learning models [54].

Lu et al. [55] proposed a framework to detect anomalies in industrial robotic sensors through a threefold process: offline training, online inference, and incremental learning. The authors used an improved GAN, named MSGAN, which utilizes an adaptive update strategy to generate high-quality synthetic anomaly samples on real robotic sensor data. On a similar approach, RoBiGAN Schnell et al. [56], a novel bidirectional GAN utilizing Wasserstein loss, for online anomaly detection and localization in robotic systems. Designed for internal robot sensor data, RoBiGAN was trained using unsupervised learning on minimal non-anomalous data.

Moreover, combining VAEs with traditional machine learning methods has led to innovations in detecting unusual patterns in data accumulated over time from various sources. An example is GenDeX, which develops a form of memory that aids in understanding new data as the characteristics of this data evolve [6]. This technique not only enhances anomaly detection but also adjusts to dynamic changes in data, improving the robustness of the monitoring systems.

2) AUTONOMY

The development and refinement of controllers for autonomous systems in diverse and dynamic environments, particularly for managing edge cases, are essential for advancing the reliability of these systems. Edge cases present significant challenges because traditional controllers perform well in controlled settings and often fail in unanticipated scenarios not included in their initial training data. GenAI plays a critical role in bridging these gaps by automating the creation of synthetic data that reflects a range of operational conditions. Moreover, research into LLMs focuses on how these models can better understand and adjust to new environments, as detailed in a survey by Wang et al. [58].

For example, integrating LLMs with conventional autonomous navigation algorithms in projects like DriveLLM has shown promise in improving reasoning and decision-making in response to edge cases [57]. This integration helps to foster proactive decision-making in unforeseen circumstances. Additionally, LLMs enhance the interaction between humans and drones through natural

TABLE 3. Comparative analysis of GenAI techniques in individual autonomous systems.

Category	Techniques Used	Key Contributions	Specific Outcomes and References
Anomaly Detection	VAEs[6], GANs[55, 56]	Enhance fault detection, identify early warning signs	Improved anomaly detection adapting to data changes, facilitating predictive maintenance
Autonomy	LLMs[57], ML Algorithms[58]	Address edge cases, enhance decision-making	Better adaptation to new environments, improved communication via natural language interfaces
Adaptability	LLM[59]	Translate control policies across different robotic arms	Control policies outperforming individual optimized controls, enhancing operational flexibility
Predictive Maintenance	GANs[50, 51], ML Algorithms[60, 61]	Predict component failures, optimize maintenance schedules	Creation of synthetic data sets for better failure prediction and maintenance planning
Fault Management	LSTMs[62, 63], GANs [60],	Improve fault detection and power efficiency	Continuous learning on sensor data, reduced power consumption, and increased processing speed

language interfaces [64], though challenges such as biases, inaccuracies, and the propensity for hallucinations in large language models call for rigorous safeguards.

Cui et al. [65] conducted a systematic review of the integration of LLMs and Vision Foundation Models (VFM) into multimodal AI systems, with an emphasis on autonomous driving applications. Beginning with an overview of MLLMs and the development of autonomous driving technology, the survey evaluates tools, datasets, and benchmarks that incorporate MLLMs in driving, transportation, and mapping systems. It underscores the contributions of recent research in the field, which addressed the role of LLMs in this sector and suggested new datasets and algorithmic enhancements. The paper identifies principal challenges and opportunities for advancing MLLMs in autonomous driving, aiming to steer further research and development efforts.

Microsoft Research has also made notable strides in integrating ChatGPT with robotics, striving to render robot control more intuitive through natural language processing. Their innovations enable ChatGPT to comprehend and execute tasks within physical settings, thus facilitating simpler human-robot interactions and eliminating the need for intricate programming abilities. Their approach utilizes specialized prompting structures and high-level APIs for language models to manage robotics tasks. Demonstrations from their work show ChatGPT conducting various operations, such as managing drones and robotic arms via straightforward commands and feedback, highlighting the crucial role of safety and simulation tests before real-world application [66].

3) ADAPTABILITY

Adaptability is essential for effectively deploying autonomous systems software across diverse makes, models, and configurations. Controllers for these systems are generally designed for specific models and configurations, which presents challenges when integrating upgraded components or new models. A notable innovation in this area is the RTX, a robotic control LLM that adeptly translates control policies across various robotic arms without requiring reconfiguring control algorithms for each new hardware piece. In some cases, these translated control policies have demonstrated superior performance to the optimal controls originally devised for

individual arms, leveraging the collective experience from multiple controllers [59].

Earlier versions of LLMs, such as GPT-3.5, were extensively trained on large text corpora from the internet but lacked practical exposure to the diverse configurations of robots and other autonomous systems. Current research addresses this gap by developing methods that constrain the models to feasible and appropriate actions for different robot types, which is vital for ensuring that these models operate effectively across various platforms [67]. This line of research is instrumental in crafting a framework that improves LLMs' comprehension of specific operational procedures and their ability to apply this knowledge effectively across different models in targeted environments.

4) PREDICTIVE MAINTENANCE

Predictive maintenance is crucial in anticipating potential failures in drone components and optimizing maintenance and component replacement schedules. This strategy involves analyzing sensor and operational data to detect early wear and tear signs, enabling timely maintenance and parts replacement interventions. Effective predictive maintenance algorithms are essential for preventing unforeseen failures during operations and reducing costs by avoiding unnecessary maintenance or replacements based on rigid schedules that do not reflect the actual condition of the components.

At the heart of predictive maintenance are traditional ML algorithms that employ metrics like Remaining Useful Life (RUL) and Health Indicators to identify motor anomalies. These algorithms' performance enhancements are often realized through synthetic data created using GANs and other GenAI methods. Moreover, various machine learning techniques, including GenAI algorithms, are refining fault diagnosis and improving predictive maintenance processes, as reported in the literature [61]. For example, GANs have been successfully used to analyze acoustic emission signals from machinery, enabling the identification and prediction of faults previously undetectable with earlier methods [50]. These networks are also crucial in creating synthetic datasets for monitoring data, which help train other ML algorithms to predict failures more accurately and optimize maintenance schedules [60]. Specifically, the GAN-FP model, a tailored

GAN for failure prediction, is instrumental in generating, balancing, and labeling training data, thereby significantly improving the effectiveness of other ML algorithms trained with this data [51].

5) FAULT MANAGEMENT

In autonomous systems, the ability to identify faults, make dynamic adjustments, and ensure safe landings is crucial. GenAI models have significantly transformed fault detection capabilities by enhancing data processing. For instance, VAEs are utilized to compress operational data into more efficient formats that LSTMs can use to improve fault detection accuracy [63]. Spatio-temporal transformer networks have effectively captured trends across different time scales, enhancing battery fault diagnosis and failure prognosis, thus bolstering predictive maintenance strategies for UAVs. An example is BERTery, which has demonstrated the ability to detect subtle changes in battery behavior that might indicate an imminent failure up to 24 hours in advance, outperforming previous machine learning techniques [68].

Generative Adversarial Networks (GAN) models are also employed to generate training samples and construct inference networks that monitor aircraft engine data, improving failure prediction capabilities [60]. Furthermore, the integration of VAE and LSTMs has proven particularly effective for continuous learning on vehicle sensor data, with Sadhu et al. achieving high accuracy rates in fault detection and classification by training on synthetic data [62]. When implementing these algorithms on cost-effective hardware, considerations such as power consumption and computational speed become significant. Porting these algorithms to Field-Programmable Gate Arrays (FPGAs), which are more energy-efficient than Graphics Processing Units (GPUs), has resulted in a fortyfold increase in processing speed and a 50% reduction in power consumption.

In the event of issues with a drone or its communication network, ensuring a safe landing to prevent secondary damage is vital. Monte Carlo algorithms have been used to calculate safety levels for potential landing zones, facilitating better decision-making under fault conditions [69]. Looking ahead, GenAI techniques like transformers may enable self-healing mechanisms in response to hardware failures, software bugs, or network disruptions. Khlaisamniang et al. have proposed a framework employing GenAI for anomaly detection, code generation, debugging, and report creation in computer systems, opening new research avenues to enhance the reliability and autonomy of autonomous systems [70].

B. FLEET

Generative Artificial Intelligence (GenAI) plays a crucial role in enhancing swarm intelligence, coordination, and the robustness of communication networks in autonomous fleets. This involves leveraging zero-trust principles for security, safety, and resilience, particularly in protecting autonomous fleets. Such approaches improve the integrity of shared

sensing, enhance coordination, and mitigate the impact of a compromised entity, such as a drone, within a fleet. As summarized in Table 4, the main contributions of GenAI in swarm operations span from enhanced data fusion and situational awareness to improved network reliability and communication efficiency.

1) SWARM INTELLIGENCE

Swarm intelligence in drone operations presents a significant opportunity to enhance the trustworthy fusion of sensor data from multiple units within a swarm. This capability is crucial in large-scale disaster response scenarios, such as flood or fire surveillance, where drones must collect and synthesize data to view the affected areas comprehensively. A notable survey on swarm intelligence by Zhou et al. [71] organizes UAV technology into five key layers: decision-making, path planning, control, communication, and application. This study elucidates the interdependencies between these layers and identifies promising research avenues for each.

For example, Generative Knowledge-Supported Transformers (GKST) has shown potential in fusing imagery from diverse perspectives, essential for interpreting moving imagery like that from drones [72]. This method could be further refined to improve the interpretation of imagery collected from various angles by a drone fleet. Different techniques for representing real-world data can be optimized for specific tasks. For instance, similarity classification algorithms sort object features in imagery, while semantic categorization algorithms classify objects into defined categories or classes. The SA-SIAM model demonstrates this by sharing information across distinct neural networks, each trained on either Semantic or Appearance data, utilizing an attention mechanism to tailor interpretation based on target and contextual information [52].

Moreover, CGANs have been utilized to predict the motion of objects relative to the changing orientations of UAVs, showing potential for enhanced motion understanding and prediction in dynamic environments [73]. This work aligns with advances in “social pooling” layers, originally designed to model interactions among nearby pedestrians for autonomous path planning. Such technology could be adapted for drones to predict the positions and movements of nearby units within a swarm or even identify and strategize against non-affiliated or adversarial drones [74].

Additionally, text-to-image diffusion models have been used to generate realistic images of UAVs in various scenarios, improving the accuracy of UAV detection algorithms by 12%. Moreover, the integration of normalizing flow models with transformers has enhanced state estimation tasks, crucial for autonomous systems to assess their relative state to other active agents, such as another drone or a vehicle [75]. These advancements highlight the potential for further research into 3D situation synthesis from the inputs of a drone fleet, indicating a promising direction for advancing the capabilities of drone swarm intelligence.

2) SWARM COORDINATION

Significant advancements in GenAI have improved task allocation across drone fleets, enhancing control in dynamic and complex environments. This represents a notable improvement over traditional fixed-algorithm approaches [76]. Chung et al. [77] detail various algorithms that boost drone communication and coordination to achieve swarm objectives. These algorithms are often structured hierarchically, enhancing autonomy at multiple levels and reducing the need for human oversight, focusing on refining control and estimation capabilities at individual and fleet-wide scales.

One innovative approach, CLIPSwarm, generates drone swarm formations based on textual descriptions [78]. This process starts with a keyword expanded into a text prompt that iteratively shapes the drone formation to visually represent the word. Utilizing the CLIP model, the algorithm encodes text and images into vectors, compares their similarity, and progressively refines the formation. CLIPSwarm is showcased through photorealistic simulations of drone shows, where formations adapt to various shapes.

Pioneering work by TII researchers has introduced multi-agent LLMs that operate on individual devices, collaborating to plan and execute tasks. These LLMs support the development of intent-based networking, translating high-level goals like reducing network energy consumption into practical, lower-level tasks. Future research aims to develop telecom-specific LLMs, enhancing self-replication in agents for improved on-device resource management, and advance consensus mechanisms. There is also an emphasis on devising methods to encode multi-modal data into more manageable formats for resource-constrained devices [79].

Additional innovations include MAPPO-GAIL, which combines Multi-Agent Proximal Policy Optimization with Generative Adversarial Imitation Learning (GAIL). This approach refines sample trajectories and policy models across the fleet, achieving a 73.3% increase in cooperative search efficiency compared to traditional Deep Reinforcement Learning (DRL) methods [72]. Moreover, the AI-Generated Optical Decision (AGOD) algorithm utilizes a diffusion model to dynamically adapt task allocation in response to evolving scenarios and human instructions, benefiting from a Deep Diffusion Soft Actor-Critic (D2SAC) algorithm that shows incremental improvements in task completion and utility metrics over standard Soft Actor-Critic (SAC) approaches [80].

Diffusion models are also proposed within a Semantic Communication framework, based on the You Only Look Once (YOLO) architecture, to reduce costs and enhance the quality of wireless transmissions needed for creating digital twins [81]. Additionally, blockchain technologies are being explored for privacy and enhanced coordination within drone fleets using Delegated Proof of Stake (DPOS) and Practical Byzantine Fault Tolerance (PBFT) mechanisms, significantly bolstering resilience against node failures and

security breaches. This blockchain approach can efficiently scale to 500 nodes with minimal overhead while effectively mitigating spoofing, denial of service, and tampering threats [82].

Furthermore, GenAI enables drones to navigate complex environments through human-like learning processes. The Generative Relation and Intention Network (GRIN) combines graph neural networks with an intention model from transformers, allowing drones to consider their relationships with others in the swarm [83]. Integrating deep reinforcement learning with transformers has also proven effective in guiding drones for regional coverage in search and rescue operations [84]. These advancements underline the growing reliance on GenAI to improve swarm intelligence, particularly in allocating tasks across fleets. GenAI promises enhanced operational efficiency and strategic autonomy in complex, variable environments.

3) COMMUNICATION RESILIENCE

Enhancing the resilience of swarm communication in diverse environments and under adversarial conditions is a significant challenge. Recent surveys on GenAI, particularly in space-ground communications, highlight potential applications in areas such as channel modeling, state information estimation, resource allocation, and network deployment. These applications also extend to semantic communications, image extraction processing, and improvements in security and privacy [85].

Generative Adversarial Networks (GAN) and Variational Autoencoders (VAE) are especially promising due to their ability to leverage real-time information and analyze radio wave propagation in dynamic settings. For example, GANs can simulate various communications scenarios to assess efficiency, while VAEs transform raw data into more effective representations for network deployments, aiding in identifying optimal patterns for deploying adaptive networks [86]. Moreover, Transformers enhance decision-making concerning network topologies, node placement, and resource allocation [87].

In Semantic Communication (SemCom), diffusion models effectively parse the semantic structure within communication streams, enhancing understanding of complex data. Transformers are instrumental in identifying and mapping relationships between distinct environmental features such as buildings and natural landscapes. GANs also offer significant potential in simulating and evaluating network configurations. This capability is particularly valuable when UAVs act as mobile base stations during emergencies, optimizing network coverage and response efficiency [88]. SemCom can greatly benefit from applying various GenAI algorithms, excelling at extracting and refining semantic information tailored to specific analytical goals. Such advancements not only enhance the quality of data available for model training but also facilitate the creation of comprehensive knowledge bases and the improvement of resource allocation strategies [89].

The physical layer of communication systems also benefits from GenAI, improving confidentiality, authentication, availability, resilience, and integrity. GenAI algorithms improve representations for complex data distributions, enhance encrypted data processing, and augment cyber-attack anomaly detection [90]. Sun et al. [91] have demonstrated the integration of GenAI in UAV communications, proposing a framework that enhances system performance, security, and efficiency through spectrum map estimation and transmission rate optimization applications.

Looking ahead, the development of Multiple Input, Multiple Output (MIMO) antennas could significantly enhance communication with autonomous systems by dynamically adjusting signals across an array of antennas. This advancement relies on models that estimate RF signal characteristics and their environmental interactions, with promising research using generative models like Variational Recurrent Neural Networks (VRNNs) and score-based approaches [92]. These innovations collectively mark a progressive step towards robust, adaptable, and secure communication frameworks for drone swarms and other autonomous systems.

4) CONSENSUS

The pursuit of automating the creation of more efficient and reliable fault-tolerant consensus protocols presents a significant opportunity in distributed computing. Vaz et al. [93] have investigated the potential of various GenAI tools to facilitate the development of these protocols, which are traditionally labor-intensive and manual. Their research highlights the limitations encountered when using AI-driven tools like ChatGPT and GitHub Co-Pilot, which were found ineffective in generating practical code for consensus protocols. However, they achieved more promising results by applying a reinforcement learning algorithm to generate candidate codes for consensus protocols. This approach included an evaluation phase to assess the suitability of the generated protocols. Although still in the preliminary stages, this research marks a significant advancement, potentially revolutionizing the development process for consensus protocols in distributed computing. It also suggests broader implications for enhancing communication and coordination within autonomous fleets, indicating a fertile area for future research and development.

C. HUMAN-ROBOT INTERACTION

Researchers are using GenAI to improve the design, development, and operation of trustworthy, resilient autonomous systems. These algorithms are pivotal in generating new data and synthesizing and distilling critical insights from the vast amounts of data captured from drone fleets, autonomous vehicles, Internet of Things (IoT) devices, and embedded systems. Obrenovic et al. [94] examine the impact of GenAI on human-robot interaction (HRI), emphasizing the ethical and societal challenges posed by increasingly human-like AI models such as ChatGPT. Obrenovic et al. highlights the need for comprehensive ethical frameworks to address

emerging issues like privacy, bias, and accountability, and calls for further research into long-term user adaptation and the interpretability of AI systems to ensure their responsible integration into society. Hunt et al. [95] discuss how LLMs are revolutionizing robotics by enhancing multimodal robot controllers that handle diverse information types, including text and audiovisual data. The survey categorizes various language applications in robotics, such as direct human-to-robot communication, robot-to-human interaction, and inter-robot dialogue, emphasizing how these capabilities can improve coordination and operational efficiency.

1) MISSION PLANNING AND EXECUTION

GenAI is set to significantly enhance mission planning and execution, facilitating seamless integration between human teams and autonomous systems such as drones. LLMs are particularly transformative, capable of translating broad mission objectives into specific, executable tasks for drone fleets. The U.S. military's exploration of this technology through Global Information Dominance Experiments (GIDE) illustrates the potential of GenAI to revolutionize operations, although details about progress are largely undisclosed, with only general updates on model advancements available [96]. Concurrently, the U.S. Small Business Innovation Research (SBIR) and Small Business Technology Transfer programs are actively soliciting proposals for reliable GenAI systems. These systems are expected to organize data and provide precise insights for Command, Control, Communications, and Computer (C4) systems, enhancing command and control capabilities [97]. Additionally, companies like C3.ai are at the forefront of embedding GenAI within defense sectors, aiming to "Accelerate data-driven decisions across personnel, intelligence, operations, logistics, sustainment, and planning functions" [98]. REAL [99] integrates LLMs into autonomous aerial robots' control and mission planning frameworks for improved resilience and adaptability. The approach harnesses LLM prior knowledge and natural language processing capabilities to enable the robots to handle unforeseen scenarios by adjusting control parameters and making real-time mission-level decisions. The system operates by interpreting natural language and numerical signals onboard the robot, allowing it to adapt its behavior dynamically to achieve mission objectives under varied conditions.

Furthermore, the Army Research Labs is advancing the use of LLMs in developing Courses of Actions (COAs) through the innovative COA-GPT model [100]. This model demonstrates rapid generation of COAs and adaptability to operator feedback in a military-adapted version of StarCraft II, showcasing the potential for GenAI to bring unprecedented agility and responsiveness to simulated operational scenarios. Integrating GenAI with strategic military planning signals a new era in decision-making, promising significantly enhanced operational efficiency and effectiveness.

TABLE 4. Detailed analysis of GenAI applications in swarm operations.

Category	Techniques Used	Key Contributions	Specific Outcomes
Swarm Intelligence	GKST [72], CGANs [73], GDM [75]	Enhance data fusion, improve situational awareness	Unified view in disaster response scenarios, advanced motion prediction, and imagery interpretation enhancements
Swarm Coordination	LLMs, MAPPO-GAIL [72], AGOD[80], GAIL, GRIN [83]	Enhance task allocation and coordination	Increased efficiency in cooperative tasks, improved task allocation responsiveness, better navigation in complex environments
Communication Resilience	GANs [88], VAEs [86], Transformers [87], SemCom Models [89]	Enhance network reliability and efficiency in communication	Optimized network configurations, enhanced data processing, improved security and privacy measures
Consensus	Reinforcement Learning [93]	Automate the development of fault-tolerant consensus protocols	Improved code generation for distributed computing protocols, enhancing communication and coordination within autonomous fleets

2) HUMAN-MACHINE TEAMING

To improve collaboration between humans and machines, GenAI technologies play a significant role. The Human Machine Teaming (HMT) Paradigm advances beyond human-in-the-loop systems to promote dynamic teamwork between people and autonomous systems within a feedback loop [101]. The Monitor, Analyze, Plan, and Execute (MAPE) framework, particularly the MAPE-KMT adaptation, manages security in unpredictable environments and highlights an architecture potentially augmented by GenAI for translating human commands into detailed tasks for autonomous systems [102]. LLMs are also making strides in processing social media data to aid disaster response. The VictimFinder project, for instance, employed LLMs to extract rescue requests from social media, where they outperformed traditional models, illustrating LLMs's potential in directing drone search and rescue operations [10]. Moreover, LLM-Brain utilizes natural language and zero-shot learning for robot task management, demonstrating an ability to improve interaction and scalability across robotic applications [103].

Additionally, Zhang and Soh's research leverages LLMs to predict human behavior for Human-Robot Interaction (HRI), showing that LLMs can match the performance of specialized models in planning tasks for robots. Despite challenges with prompt sensitivity and complex reasoning, LLMs hold promise in HRI [104]. Furthermore, the Graph Neural network Multilingual text classification framework (GNoM) combines transformers and Graph Neural Networks (GNNs) to analyze multilingual social media content, suggesting further applications of GenAI in contextually rich, multi-language environments [105].

3) META-LEARNING

In the pursuit of equipping robots with the ability to learn swiftly with minimal human intervention, the study of meta-learning is crucial. An emerging application involves using LLMs to translate human-directed requirements into dynamic, robot-specific code and policies. Large Language Model (LLM) like PaLM are utilized for code generation and composing robotic behaviors, enabling individuals without deep technical expertise to contribute to behavioral modifications. Nevertheless, there can be a loss of context

over extended interactions when shaping simpler behaviors. The Language Model Predictive Control (LMPC) framework, particularly with PaLM 2, has been fine-tuned to teach various robot models over numerous tasks, enhancing teaching effectiveness by 26.9% for novel tasks and reducing the need for human intervention while also improving adaptability to new robot configurations by 31.5% [106].

Furthermore, the MetaMorph framework leverages transformers to create a universal controller capable of converting textual task descriptions into executable commands for diverse modular robotic systems. It exhibits an impressive ability to generalize across unknown kinematic and dynamic variations and new morphologies and tasks. When trained across diverse autonomous systems, such transformers demonstrate the capacity to generalize over various configurations, ranging from unmanned aerial vehicles to ground transportation and material handling systems, thereby offering a significant advance in the adaptability and efficiency of robotic operations [107].

4) SAFETY

Safety in robotics is defined as the ability of systems to perform tasks effectively and reliably without causing unintended harm to humans or their environment. Both LLM and VLM are vulnerable to adversarial attacks that can subtly alter their inputs, leading to dangerous outputs. For instance, Wu et al. [108] details how minimal prompt or perceptual changes can severely impact the performance of these models. Their research provides a mathematical framework to understand these threats and demonstrates through extensive testing how slight input modifications can cause significant performance degradation. This highlights the critical need for robust safety measures in deploying LLMs/VLMs-based systems.

Further emphasizing the importance of structured safety interventions, Qi et al. [109] propose innovative safety mechanisms specifically for service robotics integrated with LLMs and Knowledge Graphs (KGs). They introduce Embodied Robotic Control Prompts (ERCPs) and Embodied Knowledge Graphs (EKGs), designed to enhance task accuracy and safety. By leveraging structured, factual knowledge and control prompts, these tools ensure that robotic actions

are both contextually appropriate and secure. Tested in real-world scenarios, these methods have been shown to comply more effectively with safety standards compared to traditional approaches, although they require substantial data and involve complex system integration.

D. CYBERSECURITY

Cybersecurity plays a critical role in enhancing the security and resilience of autonomous systems [135]. Extensive research has focused on how GenAI techniques, particularly GANs and VAEs, can improve malware detection and intrusion prevention. These studies also examine the potential for GenAI to refine policy management, threat simulation, and supply chain transparency. Initially, many of these techniques were developed for IT systems with robust computing capabilities, but advancements in model performance and embedded systems hardware are gradually extending these capabilities to autonomous systems.

Numerous surveys have explored the opportunities and trade-offs in applying GenAI techniques to cybersecurity, using GANs, adversarial networks, and transformers. Research has also delved into the security implications of AI on robotics and cyber defense [135]. Divakaran and Peddinti [136] discuss the transformative impact of LLMs on cybersecurity and safety, highlighting their versatility and effectiveness in addressing complex challenges within this domain. This focus underscores how the advent of advanced models offers new perspectives and solutions for significant and complex problems related to cybersecurity and safety, signaling a substantial shift in managing and mitigating such issues through cutting-edge AI technologies.

In the dynamic field of cybersecurity, the roles of LLMs and GenAI have increasingly diversified into defensive and offensive capacities. On the defensive side, innovations like SECGPT [137] and SecurityBERT [138] represent significant advancements. SECGPT enhances security by isolating app executions and regulating their interactions, effectively mitigating various security threats with minimal performance impact, marking a step forward for safer LLM-based applications. Similarly, SecurityBERT leverages the BERT model to detect cyber threats in IoT networks with an impressive 98.2% accuracy, utilizing advanced encoding techniques such as Privacy-Preserving Fixed-Length Encoding (PPFLE) and the Byte-level Byte-Pair Encoder (BBPE) Tokenizer. This model's compact size and rapid inference capabilities make it ideally suited for deployment on resource-constrained IoT devices, highlighting the transformative potential of LLMs in cybersecurity.

Conversely, LLMs and GenAI also empower offensive strategies. Net-GPT, an offensive chatbot, utilizes LLMs to conduct Man-in-the-Middle (MiTM) attacks by manipulating UAV communications with Ground Control Stations (GCS), demonstrating the dual-use potential of these technologies [139]. Additionally, BadGPT presents a novel backdoor attack against Reinforcement Learning (RL) fine-tuning in language models, highlighting vulnerabilities

where attackers can manipulate the generated text by compromising the reward model during fine-tuning [140]. This research unveils significant security risks in Natural Language Processing (NLP) paradigms and stresses the need for more rigorous defenses against such sophisticated cyber threats. Table 5 summarizes the main contributions of GenAI to cybersecurity.

1) PENTESTING

Integrating GenAI and LLMs is transforming cybersecurity, especially in penetration testing and vulnerability assessments. Hilario et al. [110] demonstrate how LLM, like ChatGPT 3.5, enhances penetration testing by offering real-time guidance and adapting testing environments with platforms such as VulnHub. This approach improves the efficiency and creativity of tests and highlights the risks of creating polymorphic malware and escalating cyber threats, emphasizing the need for robust security practices and responsible AI deployment. Concurrently, Fang et al. [111] show that LLMs, specifically GPT-4, can autonomously exploit real-world vulnerabilities informed by Common Vulnerabilities and Exposures (CVE) descriptions, stressing the importance of critically assessing LLM integration within cybersecurity frameworks to minimize risks and enhance security.

Further exploration into LLMs's capabilities in cybersecurity reveals their potential in refining software testing practices. Huang et al. conducted an extensive survey on the application of LLMs in fuzzing tests, demonstrating their superiority over traditional methods by improving API and code coverage and automating the fuzzing process. Meng et al. [113] also discuss using LLMs to enhance security in protocol implementations lacking machine-readable specifications. By developing CHATAFL, an LLM-guided protocol fuzzer, they achieve superior state and code coverage, uncovering previously unknown vulnerabilities. Xia et al. introduce Fuzz4All, a fuzzer utilizing LLMs for input generation and mutation, achieving superior code coverage and bug detection across multiple systems and languages.

2) MALWARE DETECTION

In the dynamic field of cybersecurity, GenAI has become instrumental in advancing malware detection capabilities, particularly in differentiating between legitimate and malicious code. This innovation addresses limitations inherent in traditional static analysis methods commonly employed by virus scanners. Notable models like CyBERT [122], MALBERT [123], and CySecBERT [124] have demonstrated significant advancements. CyBERT enhances security by fine-grained textual analysis, showing substantial improvements over traditional BERT models. Similarly, MALBERT has been tailored for Android applications, achieving remarkable accuracy on datasets such as Androzoo. CySecBERT extends these capabilities, excelling in intrinsic and extrinsic evaluations across various cybersecurity tasks.

TABLE 5. An overview of GenAI applications in cybersecurity.

Category	Techniques Used	Key Contributions	Specific Outcomes
Penetration Testing	LLMs [110, 111, 112, 113, 114]	Improve testing efficiency, automate vulnerability discovery	Real-time guidance in testing, automated discovery of vulnerabilities
Intrusion Detection	GANs [115, 116, 117], VAEs [118, 119], AE [120, 121]	Enhance detection capabilities, improve model training, and address data scarcity	More robust and efficient NIDS, effective against advanced intrusion methods
Malware Detection	Trans.[122, 124],LLMs [125], LSTM [126],GAN[127, 128]	Advance detection of malware in various formats, improve accuracy and efficiency	Higher accuracy in malware categorization, new methods for detecting polymorphic malware
Threat Intelligence	LLMs[129, 130], DL [131], hybrid systems [132]	Improve data aggregation, enhance keyphrase extraction	More effective security intelligence analysis, improved contextual understanding
Policy and Strategy	LLMs [133], GDM [134], GANZ[27]	Automate security policy creation, simulate and manage threats	Enhanced policy management, more convincing honeypots, better threat simulation

New research utilizes stacked bidirectional long short-term memory (Stacked BiLSTM) and generative GPT-2 models to detect malicious code in Portable Executable (PE) files [126]. These models, trained on assembly instructions extracted from PE files, demonstrate their effectiveness through various levels of analysis. The Document Level Analysis Model (DLAM) achieves an impressive 98% accuracy, while sentence-level models have shown diverse performance, with the highest reaching an 86% accuracy rate. Approaches like MallAGAN [9] use visualization methods for malware detection to enhance model training effectiveness and detection capabilities.

Combining insights from DeGPT [125] with those on the limits of LLMs like OpenAI’s Codex, we see a nuanced picture of the current state of AI in software reverse engineering. DeGPT’s approach showcases how tailored LLMs can significantly reduce the cognitive burden on reverse engineers, making decompiled code more accessible. However, the Codex study reveals that LLMs have yet to fully mature for zero-shot reverse engineering tasks. Together, these findings point to a future where, with the right training and mechanisms, LLMs could become invaluable tools for understanding and optimizing complex code. Nonetheless, realizing this potential will require overcoming current limitations through continued research and development.

Sliding Local Attention Mechanism (SLAM) leverages an API execution sequence semantic extraction engine to identify malware by its execution properties [141]. Residual attention mechanisms have also shown promise in outperforming traditional neural network approaches. I-MAD shows how transformers could create an interpretable malware detector that quantifies the impacts of features guiding the results [142]. A hierarchical transformer model that analyzes internal structure information has also shown some promise in improving predictions [143].

Furthermore, employing GANs with Recurrent Neural Networks (RNNs) for malware detection using firewall log messages treated as time-series data has shown promise [127]. Kulyadi et al. approach learns firewall operational norms and detects potential malware through deviations in established patterns. Time-series reconstruction with cosine similarity for error calculation and two encoding

methods tested demonstrate the model’s robustness. Continuous updates to the training dataset and model weights are recommended to maintain accuracy in changing network conditions. Balancing datasets remains a crucial challenge in machine learning.

3) INTRUSION DETECTION

Network Intrusion Detection System (NIDS) are pivotal in safeguarding critical digital infrastructures against evolving cyber threats. Traditional machine learning models in NIDS face challenges with advanced intrusion methods and limited diverse training samples [144]. Dunmore et al. explore the application of GANs in Intrusion Detection System (IDS), particularly for addressing dataset imbalances in attack classifications. Operational GAN mechanics, model varieties, applicable datasets, evaluation metrics, and their effectiveness across diverse IDS settings are detailed. GAN generate synthetic network traffic resembling real-world patterns, mitigating data scarcity. These synthetic attack data representing critical properties aid in developing algorithms that use various ML techniques, such as FlowGAN [145], for encrypted traffic scenarios.

Applications of GAN within the IDS can be seen on different applications. Three distinct GAN models—Vanilla GAN, Wasserstein Generative Adversarial Network (WGAN), and Conditional Tabular GAN—are utilized to generate network traffic depicting anomalous activities [116]. The results affirm that integrating GANs into NIDS significantly enhances detection capabilities for attack vectors with limited training data. GIDS, designed for automobile networks, converts Controller Area Network (CAN) bus messages into images for processing. Training solely on standard operational data, GIDS effectively identifies novel attacks, demonstrating its efficacy in real-time vehicle network applications while addressing known and unknown threats. SGAN-IDS challenges ML-NIDS robustness by generating adversarial attack flows to evade detection [117]. SGAN-IDS integrates GANs with self-attention mechanisms to create synthetic adversarial flows, targeting the vulnerability of IDSs to zero-day and adversarial attacks. The deployment of GAN-based IDS on Mobile Edge Computing (MEC) servers illustrates a strategic shift, offloading processing

from IoT devices to more capable computing hubs, thereby enhancing security functionalities across autonomous systems [146]. One intriguing use of GAN is generating more realistic-looking network traffic that is difficult for traditional IDS to detect. For example, WGAN has been shown to generate malicious traffic such that the detection using traditional IDS drops from 97.3% to 47.6% [147].

Integration of GenAI techniques such as GANs and VAEs, like in the AE-CGAN model, shows enhanced results. This model combines an AE and a CGAN to better manage skewed data distributions. The effectiveness of AE-CGAN is demonstrated using the CICIDS2017 dataset, highlighting its performance in environments with data imbalances [121]. Considerable research underscores the potential of GenAI techniques in refining models to identify key characteristics of malicious network traffic. For instance, VAEs transform raw IDS data into a compact representation, emphasizing intrusion-associated features. Nonetheless, improvements are noted with minimal labeled data, as in conditional VAEs that refine the encoding process with some labeled data, significantly enhancing detection over conventional methods [118]. Another effort combined VAEs and GANs to refine the reconstruction error of the Wasserstein distance to improve classification accuracy dramatically [119]. Additionally, the Adversarially Regularized Convolutional Autoencoder for unsupervised network anomaly Detection (ARCADE) employs a convolutional autoencoder to profile normal traffic using a subset of packet data to detect attacks efficiently. With minimal parameters, ARCADE achieves higher accuracy and quicker response times, swiftly processing initial packets in a network flow [120].

4) THREAT INTELLIGENCE

There is significant potential for enhancing the detection of emerging threats across various channels through the capabilities of LLM. LLMs are particularly adept at aggregating unstructured data from diverse sources, yet there has been limited research on their application in integrating threat intelligence. Beyond general GenAI research, specific initiatives such as CyberTwitter have demonstrated the effectiveness of using natural language processing techniques to collect and analyze data from social platforms like Twitter [132]. Mittal et al. employed Semantic Web RDF to construct a knowledge graph from Twitter data, providing security researchers with powerful tools for analyzing cybersecurity intelligence.

Similarly, Bose et al. [131] addressed the contextual limitations of existing deep learning-based and heuristic keyphrase extraction tools in the cybersecurity domain. They introduced a novel hybrid system that combines state-of-the-art (SOTA) transformers with innovative part-of-speech (POS) and role-aware tagging rules to produce detailed tag sequences from concise text corpora. This system was fine-tuned using multiple SOTA DL language model architectures. Comparative analysis revealed that this hybrid approach resulted in substantial predictive improvements,

achieving F1 scores at least 25% higher than traditional hybrid SOTA transformers on general cybersecurity datasets, with only minimal trade-offs on specialized vulnerability datasets.

Shafee et al. evaluates the performance of various LLMs such as ChatGPT, GPT4all, Dolly, Stanford Alpaca, Alpaca-LoRA, Falcon, and Vicuna in cybersecurity tasks using Twitter data. The study focuses on how these models handle binary classification and Named Entity Recognition (NER) tasks. While these LLMs excel in binary classification, matching the effectiveness of specialized models, they show limitations in NER tasks. This underscores the need for improvements in utilizing Twitter data for real-time Cyber Threat Intelligence (CTI) detection and enhancing NER accuracy. The paper recommends future research focus on refining the integration of LLM chatbots with Twitter data for cost-effective CTI on social media and improving their NER capabilities through insights from cybersecurity experts [129].

Sufi explores the use of GPT in cyber threat analysis, highlighting their capability to extract complex features from textual data on cyber incidents. The framework enables cyber strategists to query historical data in plain English and receive AI-driven insights through NLP. The validation process involved analyzing a broad set of cyber incidents, which facilitated the identification of critical elements such as Actor Type and Attack Source. However, the study also notes limitations of GPT models, including their tendency to generate non-factual content, “hallucination”, and biased text “sycophantic behavior”. These issues highlight the ongoing need for research to enhance the accuracy and reliability of AI-driven language generation in critical applications [130].

5) GenAI AND ZERO-TRUST STRATEGIES

Zero-Trust is a security model that operates on the principle of “never trust, always verify”, requiring strict verification of all users and devices, both inside and outside the network, before granting access. This approach employs least privilege access and continuous monitoring to prevent unauthorized access and ensure data security [148]. Zero-trust frameworks are increasingly crucial in boosting software supply chain visibility, particularly in critical infrastructures such as power grids. These frameworks ensure the trustworthiness of equipment and maintain rigorous control over message integrity. Continuous validation within these frameworks is key to detecting and mitigating sophisticated replay and protocol-type attacks, preserving the security and integrity of the supply chain. Collectively, these strategies highlight the transformative impact of GenAI and zero-trust frameworks in providing robust cybersecurity solutions across IT systems, network defenses, and essential supply chains [27].

Integrating GenAI into policy management, threat simulation, and software supply chain visibility is pivotal in enhancing cybersecurity across various sectors. In policy management, LLMs and GDM translate natural language descriptions into formalized rules. This automation helps

safeguard against sophisticated cyberattacks and reduces energy consumption during critical operations such as combating data poisoning attacks [134]. An example of this application is seen in Kiteworks' advocacy for resilient AI systems that incorporate zero-trust principles, ensuring rigorous validation at every data processing stage [149].

In threat simulation, LLMs are instrumental in crafting more convincing honeypots that act as decoys, diverting attackers away from critical systems. Innovative methods like Intell-dragonfly [133] generate synthetic attack patterns and scenarios. These new approaches help evade traditional IDS and enhance the training of AI classifiers. The improved classifiers are better equipped to recognize and neutralize novel attack types, thus increasing both accuracy and diversity in cyber defense mechanisms.

V. CHALLENGES

Despite the rapid advancements in GenAI, integrating these technologies into cybersecurity systems to enhance security, safety, and resilience presents substantial challenges. Initial deployments are likely to focus on low-risk tasks and optimizing ML algorithm development, potentially leading to more efficient and effective solutions.

Cost and Compute: Observations in the industry highlight that despite substantial investments surpassing \$100 billion in autonomous vehicles, these systems continue to struggle with edge cases and frequently require human intervention [150]. It is suggested that LLMs and other GenAI technologies may face similar challenges with edge cases, limiting their suitability for critical applications shortly. Furthermore, the operational costs are considerable; for example, using GPT-4 costs about \$0.006 per 750 words, and generating an image with DALL-E 2 is approximately \$0.18 per image [151]. Transformer-based LLMs, fundamental components of GenAI, require significant computational resources, which restricts their deployment to high-end servers capable of real-time integration with autonomous systems. However, hardware advancements are gradually enabling these powerful models to be run directly on autonomous systems, with FPGAs showing potential for executing GenAI models more efficiently than traditional GPUs or CPUs [62].

Adaptation and Accuracy: AI models typically need fine-tuning to maintain accuracy with historical data, and their performance may degrade under changing environmental or operational conditions. Adversaries might also adapt their strategies to bypass security measures. Continuous model updates and innovative approaches like online learning, potentially utilizing GANs, are promising for keeping models accurate over time [152]. The propensity of GenAI models to produce hallucinations requires additional measures in critical applications to ensure reliability and safety.

Privacy and Security Issues: As drones and other autonomous systems collect more data on individuals and private properties, privacy concerns grow. Security enclaves, federated learning, and homomorphic encryption

can enhance privacy while maintaining system performance. Recent advances in GenAI have led to methods that automatically mask sensitive data, such as Auto-Driving GAN (ADGAN), which focuses on relevant traffic data while obscuring backgrounds, and TrajGANs, which create synthetic datasets mimicking human movements without compromising individual privacy [153], [154]. However, the risk of exposing personal information remains, especially when models are trained on potentially identifiable data [155].

Risk and Vulnerabilities of GenAI: Security concerns specific to GenAI have garnered significant attention. The workshop proceedings by Christodorescu et al. [156] analyze these risks, further complemented by a sociotechnical safety survey from the Google DeepMind team [157]. Additionally, LLMs pose substantial privacy risks through data extraction attacks; when trained on sensitive or confidential data, these models may inadvertently leak such information, thus compromising the contextual integrity of the data [158]. Shayegani et al. [159] categorize various adversarial attacks on LLMs:

- i) *Unimodal Attacks:* Jailbreak attacks exploit the model's safety and ethical guidelines, which are designed to prevent the generation of harmful content from producing prohibited outputs. Jin et al. [160] further explore these attacks, detailing methods like gradient-based, evolutionary-based, and demonstration-based attacks for LLMs, as well as complex scenarios involving multi-agent systems and prompt-to-image manipulations for VLMs. Prompt Injection attacks manipulate inputs to elicit specific, often malicious responses, often crafting seemingly normal prompts to trigger unintended model behaviors.
- ii) *Multimodal Attacks:* These exploit vulnerabilities in LLMs that process diverse input types, such as text and images. This category includes:
 - **Input Manipulation:** Attackers may alter each input type to exploit the model's integration mechanisms, leading to inappropriate or harmful outputs.
 - **Cross-Modal Exploits:** By misaligning textual and visual data, attackers can confuse or manipulate the model to produce specific outputs.
 - **Contextual Confusion:** Attackers create ambiguous or conflicting inputs across different modalities to test the model's ability to maintain coherence and resolve ambiguities.
- iii) *Complex System Attacks:* These delve into more intricate configurations like federated learning and multi-agent systems, where vulnerabilities might be exploited to undermine the entire system or its aggregated outputs.

Ongoing robust security measures and continuous research are imperative to safeguard GenAI systems against these

evolving threats and to preserve the integrity of generated content [12].

Size and Scalability: Deploying sophisticated GenAI models, especially LLMs, on edge devices is challenging due to their size and complexity. Smaller, domain-specific models designed for particular tasks may offer a solution. For example, Microsoft's Phi-3 explores running reduced models on smartphones, with the phi-3-mini model demonstrating good understanding and reasoning capabilities, although it has limited factual knowledge and performance on complex tasks [161]. Alternatively, models could operate on nearby MEC servers or on specialized hardware designed to support larger GenAI models [146].

While GenAI offers transformative potential, its integration into critical systems requires careful consideration of numerous factors, including costs, computing requirements, adaptability, ethical implications, alignment, privacy, accuracy, size, and security. Ongoing research and development will be critical in overcoming these challenges.

A. FUTURE DIRECTIONS

1) SECURITY ENHANCEMENTS

Future efforts should focus on developing robust and efficient methods for continuous authentication and authorization of devices, fleet components, and control units, considering the resource constraints and dynamic nature of some autonomous systems, such as drone operations. Establishing secure and resilient communication networks that facilitate the safe transmission of security-related data and policies between drones, swarms, and command centers is critical. Advanced security technologies such as encryption, secure multi-party computation, and quantum key distribution should be explored. Protecting GenAIs and LLMs against various threats is also crucial. Recommendations include an AI firewall to monitor and modify the inputs and outputs of these models to prevent attacks and enforce security policies. Tools like CYBERSECEVAL 2 CyberMetric [162], SecQA [163], and WMDP-Cyber [164] are employed to test the security knowledge of models, similar to academic exams. CyberBench [165] evaluates skills like summarization within cybersecurity contexts, and LLM4Vuln focuses on enhancing vulnerability detection in models by incorporating external cybersecurity knowledge [166]. Brown et al. [167] proposed an autonomous defensive mechanism for LLMs. Self-evaluation to combat adversarial attacks using pre-trained models to classify the safety of inputs and outputs without requiring additional fine-tuning significantly reduces the cost of implementation. The method operates in three settings: Input-Only, Output-Only, and Input-Output, where the evaluator examines the user input, generated output, or both, respectively. The robustness of the defense is further validated against complex attack scenarios like direct attacks, where adversarial suffixes target both the generator and evaluator and copy-paste attacks, which manipulate the output by appending attack suffixes.

2) EXPLAINABILITY AND TRANSPARENCY

Enhancing AI-driven security decisions' explainability, transparency, and auditability is essential. This involves providing human operators with tools to understand and verify the logic behind access control decisions and trust assessments. Developing standardized metrics for explainability and transparency and adhering to best practices in trustworthy AI are crucial [168]. The IEEE Standard on Transparency of Autonomous Systems [169] details extensive transparency requirements for various stakeholders and compliance levels, although their practical implications are still being determined.

3) DYNAMIC ADAPTATION AND ROBUSTNESS TO ADVERSARIAL ATTACKS

Methods to dynamically adapt security policies and models in response to real-time environmental changes are necessary. Designing algorithms that can autonomously adjust security measures based on evolving threats and operational conditions is vital [170]. Securing GenAI models and aligning them with security mandates is challenging due to adversarial attacks, model inversion threats, backdoors, data extraction, and algorithmic biases. Research into protective measures like differential privacy, blockchain for provenance verification, quantum-resistant algorithms, and human-guided reinforcement learning is essential. Addressing broader ethical issues such as transparency, accountability, the creation of deepfakes, and model interpretability is also crucial for building robust and trustworthy systems.

4) SCALABILITY AND ETHICAL AI INTEGRATION

Addressing scalability and efficiency issues is vital for deploying GenAI-powered secure solutions. Optimizing algorithms, architectures, and resource management is necessary to ensure the efficient operation of security frameworks without excessive computational demands [171]. As autonomous systems and GenAI technologies become more widespread, incorporating ethical principles into their development and deployment becomes imperative. Research should focus on frameworks that ensure these systems operate within societal and legal standards, developing tools for detecting and mitigating bias, ensuring fairness in AI decision-making, and maintaining user privacy. Collaborations with ethicists, sociologists, and legal experts are crucial to define these frameworks clearly. Implementing oversight mechanisms like AI ethics boards and regular ethical audits can help maintain accountability and trust in autonomous systems [172].

5) ENHANCED TRANSPARENCY MECHANISMS

Developing advanced mechanisms to clarify AI decision-making, especially in critical applications like surveillance, healthcare, and law enforcement, is important. Efforts should be made to create transparent "glass-box" frameworks, standardize the documentation of AI model behaviors, and

TABLE 6. List of used acronyms.

Acronym	Full Form	Definition
AI	Artificial Intelligence	AI simulates intelligent behavior in computing.
BERT	Bidirectional Encoder Representations from Transformers	ML technique for NLP pre-training using transformers.
CGAN	Conditional Generative Adversarial Networks	GAN variant that conditions on additional inputs.
C4	Command, Control, Communications, and Computer	Systems for military command and control.
COA	Courses of Action	Strategies to achieve objectives.
CTI	Cyber Threat Intelligence	Information on cyber threats and attacks.
DL	Deep Learning	Algorithms that extract features through layers.
DPOS	Delegated Proof of Stake	Cryptocurrency consensus via voting.
DRL	Deep Reinforcement Learning	Learning to maximize cumulative reward.
D2SAC	Deep Diffusion Soft Actor-Critic	Combines diffusion models with actor-critic methods.
FPGA	Field-Programmable Gate Array	Configurable post-manufacture integrated circuit.
GAIL	Generative Adversarial Imitation Learning	Trains policies to imitate expert behavior.
GAN	Generative Adversarial Networks	Models contesting in a game framework.
GCS	Ground Control Stations	Facilities to control drones and unmanned systems.
GDM	Generative Diffusion Model	Generates data by simulating diffusion.
GenAI	Generative Artificial Intelligence	Generate new content independent of training tasks.
GIDE	Global Information Dominance Experiments	Enhances decision-making through AI and information.
GKST	Generative Knowledge-Supported Transformers	Enhances transformers with external knowledge.
GNN	Graph Neural Network	work directly with graph structures.
GPS	Global Positioning System	Satellite-based navigation system.
GPT	Generative Pre-Trained Transformer	Produces human-like text via deep learning.
HMT	Human Machine Teaming	Cooperation between humans and machines.
HRI	Human-Robot Interaction	Studies interactions between humans and robots.
IDS	Intrusion Detection System	Monitors networks for malicious activity.
IMU	Inertial Measurement Unit	Measures velocity, orientation, and gravitational forces.
IoT	Internet of Things	Network of connected devices exchanging data.
LSTM	Long Short-Term Memory	RNN capable of learning long-term dependencies.
LMPC	Language Model Predictive Control	Uses language models for decision-making.
LLM	Large Language Model	Processes generates text based on extensive data.
MAPE	Monitor, Analyze, Plan, and Execute	Framework for security management.
MIMO	Multiple Input, Multiple Output	Increases radio link capacity via multiple antennas.
MiTM	Man-in-the-Middle	Eavesdropping attack intercepting communications.
ML	Machine Learning	Allows systems to learn from data.
MLLM	Multimodal Large Language Model	Handles various data types for tasks.
MoEs	Mixture-of-Experts	Ensemble technique with specialized models.
NFM	Normalizing Flow Models	Uses transformations to learn data distributions.
NIDS	Network Intrusion Detection System	Monitors network traffic for suspicious activity.
NIST SP 800-207	National Institute of Standards and Technology Special Publication 800-207	Outlines Zero Trust Architecture principles.
PBFT	Practical Byzantine Fault Tolerance	Security consensus mechanism in distributed systems.
RUL	Remaining Useful Life	Duration components are expected to function before repair.
SAC	Soft Actor-Critic	Optimizes trade-off between return and entropy.
SemCom	Semantic Communication	Focuses on transmitting meaning over raw data.
SML	Small Language Models	Compact models for faster processing.
SRU	Simple Recurrent Unit	Efficient data processing neural network.
SOTA	state-of-the-art	Highest level of development at a particular time.
UAV	Unmanned Aerial Vehicle	Aircraft piloted remotely or autonomously.
VAE	Variational Autoencoders	AI that encodes and decodes data, learning distributions.
VFM	Vision Foundation Model	Pre-trained on images for specific visual tasks.
VRNN	Variational Recurrent Neural Network	Combines VAEs with RNNs for complex data sequences.
WGAN	Wasserstein Generative Adversarial Network	Uses Wasserstein distance to improve GAN training.
YOLO	You Only Look Once	Real-time object detection system.

ensure the traceability of training data to avoid biases and ensure data integrity. International collaborations to define and adopt transparency standards can enhance trust and facilitate global acceptance of these technologies [173].

By addressing these challenges through dedicated research, development, and collaboration, we can fully utilize GenAI to establish a secure architecture that enhances the security and resilience of autonomous systems operations. The successful integration of these technologies will result

in a more secure, adaptable, and resilient autonomous ecosystem. A holistic, multidisciplinary, and ethics-driven approach to developing AI-driven secure solutions, where the systems are continuously verifying trust to mitigate cyber threats and operational disruptions, will ensure that advancements in GenAI and LLMs are achieved responsibly, transparently, and ethically, supporting the safe and beneficial use of autonomous systems across various sectors. Through ongoing innovation and adherence to zero-trust principles, we can build a future where AI-enhanced autonomous systems are reliable, resilient, and essential for societal progress.

VI. CONCLUSION

Integrating generative AI and secure architectures offers a revolutionary approach to safeguarding autonomous systems. GenAI equips autonomous systems with intelligent, adaptive, and context-aware security, trust, and control mechanisms. At the individual drone level, GenAI enhances self-awareness, anomaly detection, autonomy, predictive maintenance, and fault management. For multiple devices such as drone fleets, GenAI facilitates the synthesis and distillation of intelligence across swarms, securing task allocation and bolstering the robustness and efficiency of communication networks. Furthermore, LLMs can optimize the flow of communications, significantly enhancing collaborative decision-making among drones. On the human interaction front, GenAI supports implementing security principles in mission planning and execution, human-machine teaming, and strengthens the reliability of robots to achieve intended goals.

Despite the potential of GenAI, significant challenges remain for enhanced security in autonomous systems. These challenges include balancing security integration with high operational costs, managing substantial computational demands, and navigating ethical and regulatory landscapes. Critical concerns include accuracy, privacy protection, and the technical challenges of deploying advanced models on edge devices. New security threats, such as prompt injection and data leakage, have also emerged, underlining the need for continuous research and development. This research must focus on creating safe, efficient, and privacy-conscious applications of GenAI in autonomous systems. A holistic, multidisciplinary approach is essential to address these issues effectively. By merging insights from AI, cybersecurity, and robotics, we can develop GenAI applications that are not only technologically advanced but also ethically sound and socially responsible. Future research should continue to innovate while adhering to rigorous security standards and ethical guidelines, ensuring that GenAI contributes positively to the evolution of autonomous systems technologies.

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REFERENCES

- [1] L. Cao, "AI robots and humanoid AI: Review, perspectives and directions," 2024, *arXiv:2405.15775*.
- [2] A. Chowdhury, G. Karmakar, J. Kamruzzaman, A. Jolfaei, and R. Das, "Attacks on self-driving cars and their countermeasures: A survey," *IEEE Access*, vol. 8, pp. 207308–207342, 2020.
- [3] A. Chougule, V. Chamola, A. Sam, F. R. Yu, and B. Sikdar, "A comprehensive review on limitations of autonomous driving and its impact on accidents and collisions," *IEEE Open J. Veh. Technol.*, vol. 5, pp. 142–161, 2024.
- [4] B. Zolfaghari, M. Abbasmollaei, F. Hajizadeh, N. Yanai, and K. Bibak, "Secure UAV (Drone) and the great promise of AI," *ACM Comput. Surveys*, vol. 56, no. 11, pp. 1–37, Nov. 2024.
- [5] T. D. Barfoot, *State Estimation for Robotics*. Cambridge, U.K.: Cambridge Univ. Press, 2017.
- [6] G. G. González, P. Casas, and A. Fernández, "Fake it till you detect it: Continual anomaly detection in multivariate time-series using generative AI," in *Proc. IEEE Eur. Symp. Secur. Privacy Workshops*, Jul. 2023, pp. 558–566.
- [7] H. Fan and S. Anjani, "IDS-GAN: Stepping up intrusion detection method using GAN algorithm," *Int. J. Infor. Comput.*, vol. 5, no. 1, pp. 19–28, 2023.
- [8] R. Pankajakshan, S. Biswal, Y. Govindarajulu, and G. Gressel, "Mapping LLM security landscapes: A comprehensive stakeholder risk assessment proposal," 2024, *arXiv:2403.13309*.
- [9] Y. Liu, J. Li, B. Liu, X. Gao, and X. Liu, "Malware identification method based on image analysis," in *Proc. 11th Int. Conf. Inform. Tech. Med. Educ. (ITME)*, 2021, pp. 157–161.
- [10] B. Zhou, L. Zou, A. Mostafavi, B. Lin, M. Yang, N. Gharaibeh, H. Cai, J. Abedin, and D. Mandal, "VictimFinder: Harvesting rescue requests in disaster response from social media with BERT," *Comput., Environ. Urban Syst.*, vol. 95, Jul. 2022, Art. no. 101824.
- [11] R. Dube, *Large Language Models in Information Security Research: A Jan. 2024 Survey*, document RG.2.2.20107.26404, 2024.
- [12] B. Zhu, N. Mu, J. Jiao, and D. Wagner, "Generative AI security: Challenges and countermeasures," 2024, *arXiv:2402.12617*.
- [13] Y. Chen, M. Cui, D. Wang, Y. Cao, P. Yang, B. Jiang, Z. Lu, and B. Liu, "A survey of large language models for cyber threat detection," *Comput. Secur.*, vol. 145, Oct. 2024, Art. no. 104016.
- [14] G. de Jesus Coelho da Silva and C. Becker Westphall, "A survey of large language models in cybersecurity," 2024, *arXiv:2402.16968*.
- [15] F. Nourmohammadzadeh Motlagh, M. Hajizadeh, M. Majid, P. Najafi, F. Cheng, and C. Meinel, "Large language models in cybersecurity: State-of-the-art," 2024, *arXiv:2402.00891*.
- [16] Y. Yao, J. Duan, K. Xu, Y. Cai, Z. Sun, and Y. Zhang, "A survey on large language model (LLM) security and privacy: The good, the bad, and the ugly," *High-Confidence Comput.*, vol. 4, no. 2, Jun. 2024, Art. no. 100211.
- [17] M. Amine Ferrag, F. Alwahedi, A. Battah, B. Cherif, A. Mechri, and N. Tihanyi, "Generative AI and large language models for cyber security: All insights you need," 2024, *arXiv:2405.12750*.
- [18] J. Zhang, H. Bu, H. Wen, Y. Chen, L. Li, and H. Zhu, "When LLMs meet cybersecurity: A systematic literature review," 2024, *arXiv:2405.03644*.
- [19] M. Hassanin and N. Moustafa, "A comprehensive overview of large language models (LLMs) for cyber defences: Opportunities and directions," 2024, *arXiv:2405.14487*.
- [20] A. Golda, K. Mekonen, A. Pandey, A. Singh, V. Hassija, V. Chamola, and B. Sikdar, "Privacy and security concerns in generative AI: A comprehensive survey," *IEEE Access*, vol. 1, pp. 1–15, 2024.
- [21] S. Javadi, N. Saeed, and B. He, "Large language models for UAVs: Current state and pathways to the future," 2024, *arXiv:2405.01745*.
- [22] G. Liu, N. Van Huynh, H. Du, D. Thai Hoang, D. Niyato, K. Zhu, J. Kang, Z. Xiong, A. Jamalipour, and D. I. Kim, "Generative AI for unmanned vehicle swarms: Challenges, applications and opportunities," 2024, *arXiv:2402.18062*.
- [23] F. Zeng, W. Gan, Y. Wang, N. Liu, and P. S. Yu, "Large language models for robotics: A survey," 2023, *arXiv:2311.07226*.

- [24] Y. Kim, D. Kim, J. Choi, J. Park, N. Oh, and D. Park, "A survey on integration of large language models with intelligent robots," 2024, *arXiv:2404.09228*.
- [25] J. Wang, Z. Wu, Y. Li, H. Jiang, P. Shu, E. Shi, H. Hu, C. Ma, Y. Liu, X. Wang, Y. Yao, X. Liu, H. Zhao, Z. Liu, H. Dai, L. Zhao, B. Ge, X. Li, T. Liu, and S. Zhang, "Large language models for robotics: Opportunities, challenges, and perspectives," 2024, *arXiv:2401.04334*.
- [26] S. Neupane, S. Mitra, I. A. Fernandez, S. Saha, S. Mittal, J. Chen, N. Pillai, and S. Rahimi, "Security considerations in AI-robotics: A survey of current methods, challenges, and opportunities," *IEEE Access*, vol. 12, pp. 22072–22097, 2024.
- [27] M. Shirajum Munir, S. Proddatoori, M. Muralidhara, W. Saad, Z. Han, and S. Shetty, "A zero trust framework for realization and defense against generative AI attacks in power grid," 2024, *arXiv:2403.06388*.
- [28] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
- [29] M. Xu, H. Du, D. Niyato, J. Kang, Z. Xiong, S. Mao, Z. Han, A. Jamalipour, D. I. Kim, X. Shen, V. C. M. Leung, and H. V. Poor, "Unleashing the power of edge-cloud generative AI in mobile networks: A survey of AIGC services," *IEEE Commun. Surveys Tuts.*, vol. 26, no. 2, pp. 1127–1170, 2nd Quart., 2024.
- [30] T. B. Brown et al., "Language models are few-shot learners," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 1877–1901.
- [31] E. Akleman, "Deep learning," *Computer*, vol. 53, no. 9, pp. 1–17, Sep. 2020.
- [32] S. Yin, C. Fu, S. Zhao, K. Li, X. Sun, T. Xu, and E. Chen, "A survey on multimodal large language models," 2023, *arXiv:2306.13549*.
- [33] OpenAI et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.
- [34] G. Team et al., "Gemini: A family of highly capable multimodal models," 2023, *arXiv:2312.11805*.
- [35] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, and G. Lample, "LLaMA: Open and efficient foundation language models," 2023, *arXiv:2302.13971*.
- [36] (2024). *Meta Llama 3*. Accessed: Aug. 2, 2024. [Online]. Available: <https://ai.meta.com/blog/meta-llama-3/>
- [37] (2023). *Mistral AI*. Accessed: Aug. 2, 2024. [Online]. Available: <https://www.mistralai.com/>
- [38] G. Penedo, Q. Malartic, D. Hesslow, R. Cojocaru, A. Cappelli, H. Alobeidli, B. Pannier, E. Almazrouei, and J. Launay, "The Refined-Web dataset for falcon LLM: Outperforming curated corpora with web data, and web data only," 2023, *arXiv:2306.01116*.
- [39] S. Mukherjee, A. Mitra, G. Jawahar, S. Agarwal, H. Palangi, and A. Awadallah, "Orca: Progressive learning from complex explanation traces of GPT-4," 2023, *arXiv:2306.02707*.
- [40] (2024). *Claude 3 Family*. Accessed: Aug. 2, 2024. [Online]. Available: <https://www.anthropic.com/news/claude-3-family>
- [41] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.
- [42] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.
- [43] L. Yang, Z. Zhang, Y. Song, S. Hong, R. Xu, Y. Zhao, W. Zhang, B. Cui, and M.-H. Yang, "Diffusion models: A comprehensive survey of methods and applications," *ACM Comput. Surv.*, vol. 56, no. 4, pp. 1–39, Apr. 2024.
- [44] Y. Tian, L. Fan, P. Isola, H. Chang, and D. Krishnan, "Stablerep: Synthetic images from text-to-image models make strong visual representation learners," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2024, pp. 1–20.
- [45] I. Kobyzev, S. J. D. Prince, and M. A. Brubaker, "Normalizing flows: An introduction and review of current methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 11, pp. 3964–3979, Nov. 2021.
- [46] P. Kirichenko, P. Izmailov, and A. G. Wilson, "Why normalizing flows fail to detect out-of-distribution data," in *Proc. NIPS*, vol. 33, 2020, pp. 20578–20589.
- [47] M. Deitke et al., "Retrospectives on the embodied AI workshop," 2022, *arXiv:2210.06849*.
- [48] X. Wu, S. Chakraborty, R. Xian, J. Liang, T. Guan, F. Liu, B. M. Sadler, D. Manocha, and A. Singh Bedi, "Highlighting the safety concerns of deploying LLMs/VLMs in robotics," 2024, *arXiv:2402.10340*.
- [49] Y. He, S. Chai, and Z. Xu, "A novel approach for state estimation using generative adversarial network," in *Proc. IEEE Int. Conf. Syst. Man Cybern. (SMC)*, Oct. 2019, pp. 2248–2253.
- [50] Y. Wang and A. Vinogradov, "Improving the performance of convolutional GAN using history-state ensemble for unsupervised early fault detection with acoustic emission signals," *Appl. Sci.*, vol. 13, no. 5, p. 3136, Feb. 2023.
- [51] S. Zheng, A. Farahat, and C. Gupta, "Generative adversarial networks for failure prediction," in *Proc. Eur. Conf. Mach. Learn. Prin. Prac. Know. Disc. Datab*, 2020, pp. 621–637.
- [52] A. He, C. Luo, X. Tian, and W. Zeng, "A twofold Siamese network for real-time object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4834–4843.
- [53] S. Chirayil Nandakumar, D. Mitchell, M. S. Erden, D. Flynn, and T. Lim, "Anomaly detection methods in autonomous robotic missions," *Sensors*, vol. 24, no. 4, p. 1330, Feb. 2024.
- [54] X. Xia, X. Pan, N. Li, X. He, L. Ma, X. Zhang, and N. Ding, "GAN-based anomaly detection: A review," *Neurocomputing*, vol. 493, pp. 497–535, Jul. 2022.
- [55] H. Lu, M. Du, K. Qian, X. He, and K. Wang, "GAN-based data augmentation strategy for sensor anomaly detection in industrial robots," *IEEE Sensors J.*, vol. 22, no. 18, pp. 17464–17474, Sep. 2022.
- [56] T. Schnell, K. Bott, L. Puck, T. Buettner, A. Roennau, and R. Dillmann, "RoBiGAN: A bidirectional Wasserstein GAN approach for online robot fault diagnosis via internal anomaly detection," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2022, pp. 4332–4337.
- [57] Y. Cui, S. Huang, J. Zhong, Z. Liu, Y. Wang, C. Sun, B. Li, X. Wang, and A. Khajepour, "DriveLLM: Charting the path toward full autonomous driving with large language models," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 1, pp. 1450–1464, Jan. 2024.
- [58] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, and J. Wen, "A survey on large language model based autonomous agents," *Frontiers Comput. Sci.*, vol. 18, no. 6, pp. 1–26, Dec. 2024.
- [59] O. X-Embodiment Collaboration et al., "Open X-embodiment: Robotic learning datasets and RT-X models," 2023, *arXiv:2310.08864*.
- [60] Q. Fu, H. Wang, J. Zhao, and X. Yan, "A maintenance-prediction method for aircraft engines using generative adversarial networks," in *Proc. IEEE 5th Int. Conf. Comput.*, Oct. 2019, pp. 225–229.
- [61] Z. Mian, X. Deng, X. Dong, Y. Tian, T. Cao, K. Chen, and T. A. Jaber, "A literature review of fault diagnosis based on ensemble learning," *Eng. Appl. Artif. Intell.*, vol. 127, Jan. 2024, Art. no. 107357.
- [62] V. Sadhu, K. Anjum, and D. Pompili, "On-board deep-learning-based unmanned aerial vehicle fault cause detection and classification via FPGAs," *IEEE Trans. Robot.*, vol. 11, no. 1, pp. 1–13, Oct. 2023.
- [63] P. Han, A. L. Ellefsen, G. Li, F. T. Holmeset, and H. Zhang, "Fault detection with LSTM-based variational autoencoder for maritime components," *IEEE Sensors J.*, vol. 21, no. 19, pp. 21903–21912, Oct. 2021.
- [64] G. Chen, X. Yu, and L. Zhong, "TypeFly: Flying drones with large language model," 2023, *arXiv:2312.14950*.
- [65] C. Cui, Y. Ma, X. Cao, W. Ye, Y. Zhou, K. Liang, J. Lu, Z. Yang, K.-D. Liao, T. Gao, E. Li, K. Tang, Z. Cao, T. Zhou, A. Liu, X. Yan, S. Mei, J. Cao, Z. Wang, and C. Zheng, "A survey on multimodal large language models for autonomous driving," in *Proc. IEEE/CVF Winter Conf. App. Comput. Vis.*, Aug. 2024, pp. 958–979.
- [66] S. H. Vemprala, R. Bonatti, A. Buckner, and A. Kapoor, "ChatGPT for robotics: Design principles and model abilities," *IEEE Access*, vol. 12, pp. 55682–55696, 2024.
- [67] M. Ahn et al., "Do as I can, not as I say: Grounding language in robotic affordances," 2022, *arXiv:2204.01691*.
- [68] J. Zhao, X. Feng, J. Wang, Y. Lian, M. Ouyang, and A. F. Burke, "Battery fault diagnosis and failure prognosis for electric vehicles using spatio-temporal transformer networks," *Appl. Energy*, vol. 352, Dec. 2023, Art. no. 121949.
- [69] L. Tong, X. Gan, L. Yu, and H. Zhang, "Evaluation of safety target level of unmanned aerial vehicle system in fusion airspace," in *Proc. IEEE Int. Conf. Artif. Intell. Comput. Appl. (ICAICA)*, Jun. 2022, pp. 375–379.
- [70] P. Khlaisamniang, P. Khomduncan, K. Saetan, and S. Wonglaphsuwan, "Generative AI for self-healing systems," in *Proc. 18th Int. Joint Symp. Artif. Intell. Natural Lang. Process. (ISAI-NLP)*, Nov. 2023, pp. 1–6.

- [71] Y. Zhou, B. Rao, and W. Wang, "UAV swarm intelligence: Recent advances and future trends," *IEEE Access*, vol. 8, pp. 183856–183878, 2020.
- [72] X. Yu, W. Liao, C. Qu, Q. Bao, and Z. Xu, "UAV cooperative search based on multi-agent generative adversarial imitation learning," in *Proc. Int. Conf. Mach. Learn., Cloud Comput. Intell. Mining (MLCCIM)*, Aug. 2022, pp. 441–446.
- [73] H. Yu, G. Li, L. Su, B. Zhong, H. Yao, and Q. Huang, "Conditional GAN based individual and global motion fusion for multiple object tracking in UAV videos," *Pattern Recognit. Lett.*, vol. 131, pp. 219–226, Mar. 2020.
- [74] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human trajectory prediction in crowded spaces," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 961–971.
- [75] H. Delecki, L. A. Kruse, M. R. Schlichting, and M. J. Kochenderfer, "Deep normalizing flows for state estimation," in *Proc. Int. Conf. Inform. Fusion*, 2023, pp. 1–6.
- [76] G. M. Skaltsis, H.-S. Shin, and A. Tsourdos, "A survey of task allocation techniques in MAS," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, Jun. 2021, pp. 488–497.
- [77] S.-J. Chung, A. A. Paranjape, P. Dames, S. Shen, and V. Kumar, "A survey on aerial swarm robotics," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 837–855, Aug. 2018.
- [78] P. Pueyo, E. Montijano, A. C. Murillo, and M. Schwager, "CLIPSwarm: Generating drone shows from text prompts with vision-language models," 2024, *arXiv:2403.13467*.
- [79] H. Zou, Q. Zhao, L. Bariah, M. Bennis, and M. Debbah, "Wireless multi-agent generative AI: From connected intelligence to collective intelligence," 2023, *arXiv:2307.02757*.
- [80] H. Du, Z. Li, D. Niyato, J. Kang, Z. Xiong, H. Huang, and S. Mao, "Generative AI-aided optimization for AI-generated content (AIGC) services in edge networks," 2023, *arXiv:2303.13052*.
- [81] B. Du, H. Du, H. Liu, D. Niyato, P. Xin, J. Yu, M. Qi, and Y. Tang, "YOLO-based semantic communication with generative AI-aided resource allocation for digital twins construction," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 7664–7678, Mar. 2024.
- [82] S. Hafeez, R. Cheng, L. Mohjazi, Y. Sun, and M. Ali Imran, "Blockchain-enhanced UAV networks for post-disaster communication: A decentralized flocking approach," 2024, *arXiv:2403.04796*.
- [83] L. Li, J. Yao, L. Wenliang, T. He, T. Xiao, J. Yan, D. Wipf, and Z. Zhang, "Grin: Generative relation and intention network for multi-agent trajectory prediction," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 27107–27118.
- [84] D. Fuertes, C. R. del-Blanco, F. Jaureguizar, J. J. Navarro, and N. García, "Solving routing problems for multiple cooperative unmanned aerial vehicles using transformer networks," *Eng. Appl. Artif. Intell.*, vol. 122, Jun. 2023, Art. no. 106085.
- [85] R. Zhang, H. Du, D. Niyato, J. Kang, Z. Xiong, A. Jamalipour, P. Zhang, and D. In Kim, "Generative AI for space-air-ground integrated networks (SAGIN)," 2023, *arXiv:2311.06523*.
- [86] H. Du, R. Zhang, D. Niyato, J. Kang, Z. Xiong, D. I. Kim, X. S. Shen, and H. V. Poor, "Exploring collaborative distributed diffusion-based AI-generated content (AIGC) in wireless networks," *IEEE Netw.*, vol. 1, no. 1, pp. 1–8, Jun. 2023.
- [87] G. Fontanesi, A. Zhu, M. Arvaneh, and H. Ahmadi, "A transfer learning approach for UAV path design with connectivity outage constraint," *IEEE Internet Things J.*, vol. 10, no. 6, pp. 4998–5012, Mar. 2023.
- [88] M. Ružička, M. Volosin, J. Gazda, T. Maksymuk, L. Han, and M. Dohler, "Fast and computationally efficient generative adversarial network algorithm for unmanned aerial vehicle-based network coverage optimization," *Int. J. Distrib. Sensor Netw.*, vol. 18, no. 3, Mar. 2022, Art. no. 155014772210755.
- [89] C. Liang, H. Du, Y. Sun, D. Niyato, J. Kang, D. Zhao, and M. A. Imran, "Generative AI-driven semantic communication networks: Architecture, technologies and applications," 2023, *arXiv:2401.00124*.
- [90] J. Wang, H. Du, D. Niyato, J. Kang, S. Cui, X. S. Shen, and P. Zhang, "Generative AI for integrated sensing and communication: Insights from the physical layer perspective," *IEEE Wireless Commun.*, vol. 1, no. 1, pp. 1–10, Aug. 2024.
- [91] G. Sun, W. Xie, D. Niyato, H. Du, J. Kang, J. Wu, S. Sun, and P. Zhang, "Generative AI for advanced UAV networking," 2024, *arXiv:2404.10556*.
- [92] M. Arvinte and J. I. Tamir, "MIMO channel estimation using score-based generative models," *IEEE Trans. Wireless Commun.*, vol. 1, no. 2, pp. 1–16, Jun. 2022.
- [93] D. Vaz, D. R. Matos, M. L. Pardo, and M. Correia, "Automatic generation of distributed algorithms with generative AI," in *Proc. 53rd Annu. IEEE/IFIP Int. Conf. Dependable Syst. Netw. Supplemental Volume*, Jun. 2023, pp. 127–131.
- [94] B. Obrenovic, X. Gu, G. Wang, D. Godinic, and I. Jakhongirov, "Generative AI and human-robot interaction: Implications and future agenda for business, society and ethics," *AI Soc.*, vol. 1, pp. 1–14, Mar. 2024.
- [95] W. Hunt, S. D. Ramchurn, and M. D. Soorati, "A survey of language-based communication in robotics," 2024, *arXiv:2406.04086*.
- [96] J. Harper, (2024). *Pentagon Testing Generative AI in 'Global Information Dominance' Experiments*. Accessed: Aug. 2, 2024. [Online]. Available: <https://defensescoop.com/2023/07/14/pentagon-testing-generative-ai-in-global-information-dominance-experiments/>
- [97] (2024). *Trustworthy Generative Artificial Intelligence (GenAI) to Structure Data and Deliver Accurate Insights of Command, Control, Communication and Computer (C4) Systems*. Accessed: Aug. 2, 2024. [Online]. Available: <https://legacy.www.sbir.gov/node/2479917>
- [98] (2024). *Generative AI for Defense*. Accessed: Aug. 2, 2024. [Online]. Available: <https://c3.ai/generative-ai-for-defense/>
- [99] A. Tagliabue, K. Kondo, T. Zhao, M. Peterson, C. T. Tewari, and J. P. How, "REAL: Resilience and adaptation using large language models on autonomous aerial robots," 2023, *arXiv:2311.01403*.
- [100] V. G. Goecks and N. Waytowich, "COA-GPT: Generative pre-trained transformers for accelerated course of action development in military operations," in *Proc. Int. Conf. Mil. Commun. Inf. Syst. (ICMCIS)*, Apr. 2024, pp. 1–24.
- [101] C. Zhang, J. Chen, J. Li, Y. Peng, and Z. Mao, "Large language models for human-robot interaction: A review," *Biomimetic Intell. Robot.*, vol. 1, Jun. 2023, Art. no. 100131.
- [102] J. Cleland-Huang, A. Agrawal, M. Vierhauser, M. Murphy, and M. Prieto, "Extending MAPE-K to support human-machine teaming," in *Proc. 17th Symp. Softw. Eng. Adap. Self-Managing Syst.*, 2022, pp. 120–131.
- [103] J. Mai, J. Chen, B. Li, G. Qian, M. Elhoseiny, and B. Ghanem, "LLM as a robotic brain: Unifying egocentric memory and control," 2023, *arXiv:2304.09349*.
- [104] B. Zhang and H. Soh, "Large language models as zero-shot human models for human-robot interaction," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2023, pp. 7961–7968.
- [105] J. Liang et al., "Learning to learn faster from human feedback with language model predictive control," 2024, *arXiv:2402.11450*.
- [106] R. Anil et al., "PaLM 2 technical report," 2023, *arXiv:2305.10403*.
- [107] A. Gupta, L. Fan, S. Ganguli, and L. Fei-Fei, "MetaMorph: Learning universal controllers with transformers," 2022, *arXiv:2203.11931*.
- [108] X. Wu, T. Guan, D. Li, S. Huang, X. Liu, X. Wang, R. Xian, A. Shrivastava, F. Huang, J. Lee Boyd-Graber, T. Zhou, and D. Manocha, "AUTOHALLUSION: Automatic generation of hallucination benchmarks for vision-language models," 2024, *arXiv:2406.10900*.
- [109] Y. Qi, G. Kyebambo, S. Xie, W. Shen, S. Wang, B. Xie, B. He, Z. Wang, and S. Jiang, "Safety control of service robots with LLMs and embodied knowledge graphs," 2024, *arXiv:2405.17846*.
- [110] E. Hilario, S. Azam, J. Sundaram, K. Imran Mohammed, and B. Shanmugam, "Generative AI for pentesting: The good, the bad, the ugly," *Int. J. Inf. Secur.*, vol. 23, no. 3, pp. 2075–2097, Jun. 2024.
- [111] R. Fang, R. Bindu, A. Gupta, and D. Kang, "LLM agents can autonomously exploit one-day vulnerabilities," 2024, *arXiv:2404.08144*.
- [112] L. Huang, P. Zhao, H. Chen, and L. Ma, "Large language models based fuzzing techniques: A survey," 2024, *arXiv:2402.00350*.
- [113] R. Meng, M. Mirchev, M. Böhme, and A. Roychoudhury, "Large language model guided protocol fuzzing," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2024, pp. 1–15.
- [114] C. S. Xia, M. Paltenghi, J. Le Tian, M. Pradel, and L. Zhang, "Fuzz4all: Universal fuzzing with large language models," in *Proc. IEEE/ACM 46th Interface Conf. Soft. Eng.*, Jul. 2024, pp. 1–13.
- [115] A. Dunmore, J. Jang-Jaccard, F. Sabrina, and J. Kwak, "A comprehensive survey of generative adversarial networks (GANs) in cybersecurity intrusion detection," *IEEE Access*, vol. 11, pp. 76071–76094, 2023.
- [116] X. Zhao, K. Wai Fok, and V. L. L. Thing, "Enhancing network intrusion detection performance using generative adversarial networks," 2024, *arXiv:2404.07464*.

- [117] S. Aldhaheri and A. Alhuzali, "SGAN-IDS: Self-attention-based generative adversarial network against intrusion detection systems," *Sensors*, vol. 23, no. 18, p. 7796, Sep. 2023.
- [118] A. Hannan, C. Gruhl, and B. Sick, "Anomaly based resilient network intrusion detection using inferential autoencoders," in *Proc. IEEE Int. Conf. Cyber Secur. Resilience (CSR)*, Jul. 2021, pp. 1–7.
- [119] C. Park, J. Lee, Y. Kim, J.-G. Park, H. Kim, and D. Hong, "An enhanced AI-based network intrusion detection system using generative adversarial networks," *IEEE Internet Things J.*, vol. 10, no. 3, pp. 2330–2345, Feb. 2023.
- [120] W. T. Lunardi, M. A. Lopez, and J.-P. Giacalone, "ARCADE: Adversarially regularized convolutional autoencoder for network anomaly detection," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 2, pp. 1305–1318, Jul. 2022.
- [121] J. Lee and K. Park, "AE-CGAN model based high performance network intrusion detection system," *Appl. Sci.*, vol. 9, no. 20, p. 4221, Oct. 2019.
- [122] P. Ranade, A. Piplai, A. Joshi, and T. Finin, "CyBERT: Contextualized embeddings for the cybersecurity domain," in *Proc. IEEE Int. Conf. Big Data*, Dec. 2021, pp. 3334–3342.
- [123] A. Rahali and M. A. Akhloufi, "MalBERT: Using transformers for cybersecurity and malicious software detection," 2021, *arXiv:2103.03806*.
- [124] M. Bayer, P. Kuehn, R. Shanesaz, and C. Reuter, "CySecBERT : A domain-adapted language model for the cybersecurity domain," *ACM Trans. Privacy Secur.*, vol. 27, no. 2, pp. 1–20, May 2024.
- [125] P. Hu, R. Liang, and K. Chen, "DeGPT: Optimizing decompiler output with LLM," in *Proc. Neww. Distrib. Syst. Secur. Symp.*, 2024, pp. 1–27.
- [126] D. Demirci, N. Sahin, M. Sirlancis, and C. Acarturk, "Static malware detection using stacked BiLSTM and GPT-2," *IEEE Access*, vol. 10, pp. 58488–58502, 2022.
- [127] S. P. Kulyadi, P. Mohandas, S. K. S. Kumar, M. J. S. Raman, and V. S. Vasan, "Anomaly detection using generative adversarial networks on firewall log message data," in *Proc. 13th Intern. Conf. Elec. Comput. Art. Int. (ECAI)*, 2021, pp. 1–6.
- [128] H. Pearce, B. Tan, P. Krishnamurthy, F. Khorrami, R. Karri, and B. Dolan-Gavitt, "Pop quiz! Can a large language model help with reverse engineering?" 2022, *arXiv:2202.01142*.
- [129] S. Shafee, A. Bessani, and P. M. Ferreira, "Evaluation of LLM chatbots for OSINT-based cyber threat awareness," 2024, *arXiv:2401.15127*.
- [130] F. Sufi, "An innovative GPT-based open-source intelligence using historical cyber incident reports," *Natural Lang. Process. J.*, vol. 7, Jun. 2024, Art. no. 100074.
- [131] A. Bose, H. Yang, M. Shivers, A. Orazgeldiyev, and W. H. Hsu, "Context-augmented key phrase extraction from short texts for cyber threat intelligence tasks," in *Proc. IEEE Int. Conf. Intell. Secur. Informat. (ISI)*, Oct. 2023, pp. 1–6.
- [132] S. Mittal, P. K. Das, V. Mulwad, A. Joshi, and T. Finin, "CyberTwitter: Using Twitter to generate alerts for cybersecurity threats and vulnerabilities," in *Proc. IEEE/ACM Int. Conf. Adv. Social Netw. Anal. Mining (ASONAM)*, Aug. 2016, pp. 860–867.
- [133] X. Wu, Q. Qiu, J. Li, and Y. Zhao, "Intell-dragonfly: A cybersecurity attack surface generation engine based on artificial intelligence-generated content technology," 2023, *arXiv:2311.00240*.
- [134] H. Du, D. Niyato, J. Kang, Z. Xiong, K.-Y. Lam, Y. Fang, and Y. Li, "Spear or shield: Leveraging generative AI to tackle security threats of intelligent network services," 2023, *arXiv:2306.02384*.
- [135] S. Neupane, I. A. Fernandez, S. Mittal, and S. Rahimi, "Impacts and risk of generative AI technology on cyber defense," 2023, *arXiv:2306.13033*.
- [136] D. Mon Divakaran and S. Teja Peddinti, "LLMs for cyber security: New opportunities," 2024, *arXiv:2404.11338*.
- [137] Y. Wu, F. Roesner, T. Kohn, N. Zhang, and U. Iqbal, "SecGPT: An execution isolation architecture for LLM-based systems," 2024, *arXiv:2403.04960*.
- [138] M. A. Ferrag, M. Ndhlovu, N. Tihanyi, L. C. Cordeiro, M. Deb-bah, T. Lestable, and N. S. Thandi, "Revolutionizing cyber threat detection with large language models: A privacy-preserving BERT-based lightweight model for IoT/IIoT devices," *IEEE Access*, vol. 12, pp. 23733–23750, 2024.
- [139] B. Piggott, S. Patil, G. Feng, I. Odat, R. Mukherjee, B. Dharmalingam, and A. Liu, "Net-GPT: A LLM-empowered man-in-the-middle chatbot for unmanned aerial vehicle," in *Proc. IEEE/ACM Symp. Edge Comput. (SEC)*, Dec. 2023, pp. 287–293.
- [140] J. Shi, Y. Liu, P. Zhou, and L. Sun, "BadGPT: Exploring security vulnerabilities of ChatGPT via backdoor attacks to InstructGPT," 2023, *arXiv:2304.12298*.
- [141] J. Chen, S. Guo, X. Ma, H. Li, J. Guo, M. Chen, and Z. Pan, "SLAM: A malware detection method based on sliding local attention mechanism," *Secur. Commun. Netw.*, vol. 2020, pp. 1–11, Sep. 2020.
- [142] M. Q. Li, B. C. M. Fung, P. Charland, and S. H. H. Ding, "I-MAD: Interpretable malware detector using galaxy transformer," *Comput. Secur.*, vol. 108, Sep. 2021, Art. no. 102371.
- [143] X. Hu, R. Sun, K. Xu, Y. Zhang, and P. Chang, "Exploit internal structural information for IoT malware detection based on hierarchical transformer model," in *Proc. IEEE 19th Int. Conf. Trust, Secur. Privacy Comput. Commun. (TrustCom)*, Dec. 2020, pp. 927–934.
- [144] D. Herzalla, W. T. Lunardi, and M. Andreoni, "TII-SSRC-23 dataset: Typological exploration of diverse traffic patterns for intrusion detection," *IEEE Access*, vol. 11, pp. 118577–118594, 2023.
- [145] Z. Wang, P. Wang, X. Zhou, S. Li, and M. Zhang, "FLOWGAN: Unbalanced network encrypted traffic identification method based on GAN," in *Proc. IEEE Intl. Conf. Parallel Distrib. Process. Appl., Big Data Cloud Comput., Sustain. Comput. Commun., Social Comput. Netw. (ISPA/BDCloud/SocialCom/SustainCom)*, Dec. 2019, pp. 975–983.
- [146] H. Sedjelmaci, "Attacks detection and decision framework based on generative adversarial network approach: Case of vehicular edge computing network," *Trans. Emerg. Telecommun. Technol.*, vol. 33, no. 10, Oct. 2022, Art. no. e4073.
- [147] Q. Yan, M. Wang, W. Huang, X. Luo, and F. R. Yu, "Automatically synthesizing DoS attack traces using generative adversarial networks," *Int. J. Mach. Learn. Cybern.*, vol. 10, no. 12, pp. 3387–3396, Dec. 2019.
- [148] V. Stafford, "Zero trust architecture," *NIST special publication*, vol. 800, p. 207, Jun. 2020.
- [149] T. Freestone. (2024). *Building Trust in Generative AI With a Zero Trust Approach*. Accessed: Aug. 2, 2024. [Online]. Available: <https://www.kiteworks.com/cybersecurity-risk-management/zero-trust-generative-ai/>
- [150] G. Marcus. (2024). *The Second Worst100B Investment in The History of AI?*. Accessed: Aug. 2, 2024. [Online]. Available: <https://garymarcus.substack.com/p/the-second-worst-100b-investment?publicationid=888615>
- [151] H. Huang. (2023). *What CEOs Need to Know About the Costs of Adopting GenAI*. Accessed: Aug. 2, 2024. [Online]. Available: <https://hbr.org/2023/11/what-ceos-need-to-know-about-the-costs-of-adopting-genai>
- [152] W. L. Tan and T. Truong-Huu, "Enhancing robustness of malware detection using synthetically-adversarial samples," in *Proc. IEEE Global Communications Conference*, Aug. 2020, pp. 1–6.
- [153] Z. Xiong, Z. Cai, Q. Han, A. Alrawais, and W. Li, "ADGAN: Protect your location privacy in camera data of auto-driving vehicles," *IEEE Trans. Ind. Informat.*, vol. 17, no. 9, pp. 6200–6210, Sep. 2021.
- [154] O. Adeboye, T. Dargahi, M. Babaie, M. Saraei, and C.-M. Yu, "DeepClean: A robust deep learning technique for autonomous vehicle camera data privacy," *IEEE Access*, vol. 10, pp. 124534–124544, 2022.
- [155] M. Nasr, N. Carlini, J. Hayase, M. Jagielski, A. Feder Cooper, D. Ippolito, C. A. Choquette-Choo, E. Wallace, F. Tramèr, and K. Lee, "Scalable extraction of training data from (Production) language models," 2023, *arXiv:2311.17035*.
- [156] M. Christodorescu, R. Craven, S. Feizi, N. Gong, M. Hoffmann, S. Jha, Z. Jiang, M. S. Kamarpashti, J. Mitchell, J. Newman, E. Probasco, Y. Qi, K. Shams, and M. Turek, "Securing the future of GenAI: Policy and technology," *Cryptol. ePrint Arch.*, Paper 2024/855, 2024. [Online]. Available: <https://eprint.iacr.org/2024/855>
- [157] L. Weidinger, M. Rauh, N. Marchal, A. Manzini, L. Anne Hendricks, J. Mateos-Garcia, S. Bergman, J. Kay, C. Griffin, B. Bariach, I. Gabriel, V. Rieser, and W. Isaac, "Sociotechnical safety evaluation of generative AI systems," 2023, *arXiv:2310.11986*.
- [158] N. Carlini, J. Hayes, M. Nasr, M. Jagielski, V. Schwag, F. Tramèr, B. Balle, D. Ippolito, and E. Wallace, "Extracting training data from diffusion models," in *Proc. 32nd USENIX Secur.*, 2023, pp. 5253–5270.
- [159] E. Shayegani, M. Abdullah Al Mamun, Y. Fu, P. Zaree, Y. Dong, and N. Abu-Ghazaleh, "Survey of vulnerabilities in large language models revealed by adversarial attacks," 2023, *arXiv:2310.10844*.
- [160] H. Jin, L. Hu, X. Li, P. Zhang, C. Chen, J. Zhuang, and H. Wang, "JailbreakZoo: Survey, landscapes, and horizons in jailbreaking large language and vision-language models," 2024, *arXiv:2407.01599*.

- [161] M. Abidin et al., “Phi-3 technical report: A highly capable language model locally on your phone,” 2024, *arXiv:2404.14219*.
- [162] N. Tihanyi, M. A. Ferrag, R. Jain, T. Bisztray, and M. Debbah, “CyberMetric: A benchmark dataset based on retrieval-augmented generation for evaluating LLMs in cybersecurity knowledge,” 2024, *arXiv:2402.07688*.
- [163] Z. Liu, “SecQA: A concise question-answering dataset for evaluating large language models in computer security,” 2023, *arXiv:2312.15838*.
- [164] N. Li et al., “The WMDP benchmark: Measuring and reducing malicious use with unlearning,” 2024, *arXiv:2403.03218*.
- [165] Z. Liu, J. Shi, and J. F. Buford, “Cyberbench: A multi-task benchmark for evaluating large language models in cybersecurity,” in *Proc. Workshop Art. Inte. Cyb. Secur. (AICS)*, 2024, pp. 1–20.
- [166] Y. Sun, D. Wu, Y. Xue, H. Liu, W. Ma, L. Zhang, M. Shi, and Y. Liu, “LLM4 Vuln: A unified evaluation framework for decoupling and enhancing LLMs’ vulnerability reasoning,” 2024, *arXiv:2401.16185*.
- [167] H. Brown, L. Lin, K. Kawaguchi, and M. Shieh, “Self-evaluation as a defense against adversarial attacks on LLMs,” 2024, *arXiv:2407.03234*.
- [168] L. Methnani, M. Chiou, V. Dignum, and A. Theodorou, “Who’s in charge here? A survey on trustworthy AI in variable autonomy robotic systems,” *ACM Comput. Surveys*, vol. 56, no. 7, pp. 1–32, Jul. 2024.
- [169] *IEEE Standard for Transparency of Autonomous Systems*, Standard IEEE 7001-2021, 2022.
- [170] J. Perez-Cerrolaza, J. Abella, M. Borg, C. Donzella, J. Cerquides, F. J. Cazorla, C. Englund, M. Tauber, G. Nikolakopoulos, and J. L. Flores, “Artificial intelligence for safety-critical systems in industrial and transportation domains: A survey,” *ACM Comput. Surveys*, vol. 56, no. 7, pp. 1–40, Jul. 2024.
- [171] Y.-C. Wang, J. Xue, C. Wei, and C.-C. J. Kuo, “An overview on generative AI at scale with edge-cloud computing,” *IEEE Open J. Commun. Soc.*, vol. 1, pp. 1–27, 2023.
- [172] S. Russell, S. Hauer, R. Altman, and M. Veloso, “Ethics of artificial intelligence,” *Nature*, vol. 521, no. 7553, pp. 415–416, 2015.
- [173] A. Tang, K. Li, K. O. Kwok, L. Cao, S. Luong, and W. Tam, “The importance of transparency: Declaring the use of generative artificial intelligence (AI) in academic writing,” *J. Nursing Scholarship*, vol. 56, no. 2, pp. 314–318, Mar. 2024.



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